5G RAN

QoS Management Feature Parameter Description

Issue Draft B

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Contents

1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes

  Changes in functions and their corresponding parameters
- Editorial changes

  Improvements or revisions to the documentation

1.1 5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

Technical Changes

Change Description	Parameter Change	RAT	Base Station Model
Added an impact relationship between downlink scheduling priority weighting factors and downlink inter-gNodeB CA. For details, see **4.3.2.3 Function Impacts**.	None	Low-frequency TDD High-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Editorial Changes

Added descriptions of the impacts of and suggestions on scheduling priority weighting factor configuration. For details, see **4.2.2 Impacts** and **4.4.1.1 Data Preparation**.

Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

1.2 5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 03 (2025-08-30).

Technical Changes

Change Description	Parameter Change	RAT	Base Station Model
Added the impact relationships between flexible experience differentiation and the following functions in **4.3.2.3 Function Impacts**: • Maximum rate limitation for GBR services • Minimum rate guarantee for GBR services • Uplink UE-AMBR-based rate limitation for non-GBR services • Downlink UE-AMBR-based rate limitation for non-GBR services • Minimum uplink rate guarantee for non-GBR services • Minimum downlink rate guarantee for non-GBR services	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added support for PDU Session Resource Modify QCI mapping optimization. For details, see: • **4.1.1.2.2 Configuring 5QIs and Corresponding QoS Attributes** • **4.3.2.1 Prerequisite Functions** • **4.4.1.1 Data Preparation** • **4.4.1.2 Using MML Commands**	Modified parameter: Added the **PDU_SESSION_MOD_QCI_MAP_OPT_SW** option to the **gNodeBParam**.***CompatibilityAlgoSwitch*** parameter.	Low-frequency TDD High-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added support for downlink PDB guarantee status check in the notification control function. For details, see: • **4.1.2.3 QoS Guarantee for Non-Scheduling Procedure** • **4.2.2 Impacts** • **4.4.1.1 Data Preparation** • **4.4.1.2 Using MML Commands**	Added the **gNBQciBearer**.***QncArrangeType*** parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added a mutually exclusive relationship between **QoS Class Identifier** and QCI-specific uplink data compression. For details, see **4.3.2.2 Mutually Exclusive Functions**.	None	High-frequency TDD	5900 series base stations (macro base stations)
Added a mutually exclusive relationship between **QoS Class Identifier** and QCI-specific uplink data compression. For details, see **4.3.2.2 Mutually Exclusive Functions**.	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Editorial Changes

Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FBFD-020102	Radio QoS Management	**4 QoS Management**

2.3 Differences Between NSA and SA

Function Name	Difference	Chapter/Section
QoS management	Supported in both NSA and SA networking, with the following differences: The service-to-bearer mapping modes and QoS guarantee for uplink and downlink scheduling vary with the networking. For details, see **4 QoS Management**. Downlink PDB-based differentiated scheduling is supported only in SA networking. For details, see **4.1.2.1.7 Downlink PDB-based Differentiated Scheduling**. Notification control is supported only in SA networking. For details, see **4.1.2.3 QoS Guarantee for Non-Scheduling Procedure**.	**4 QoS Management**

2.4 Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
QoS management	Supported in both high and low frequency bands, with the following differences: Downlink PDB-based differentiated scheduling is supported only in low frequency bands. For details, see **4.1.2.1.7 Downlink PDB-based Differentiated Scheduling**. Notification control is supported only in low frequency bands. For details, see **4.1.2.3 QoS Guarantee for Non-Scheduling Procedure**.	**4 QoS Management**

3 Overview

Quality of service (QoS) is a measure of the overall performance of operator-delivered services, and indicates the level of customer satisfaction. QoS management is a mechanism that helps a network meet service quality requirements. The mechanism does this by implementing end-to-end (E2E) coordination among all the involved network nodes, from the initiation to the response of a service.

gNodeB QoS management includes radio QoS management and transmission QoS management. This document describes radio QoS management. For details about transmission QoS management, see *Transmission Resource Management*.

The Radio QoS Management feature guarantees QoS for different services and UEs. This feature also enables network resource competition among different services, ensuring differentiated experience.

QoS management is supported in both non-standalone (NSA) and standalone (SA) networking.

NSA Networking

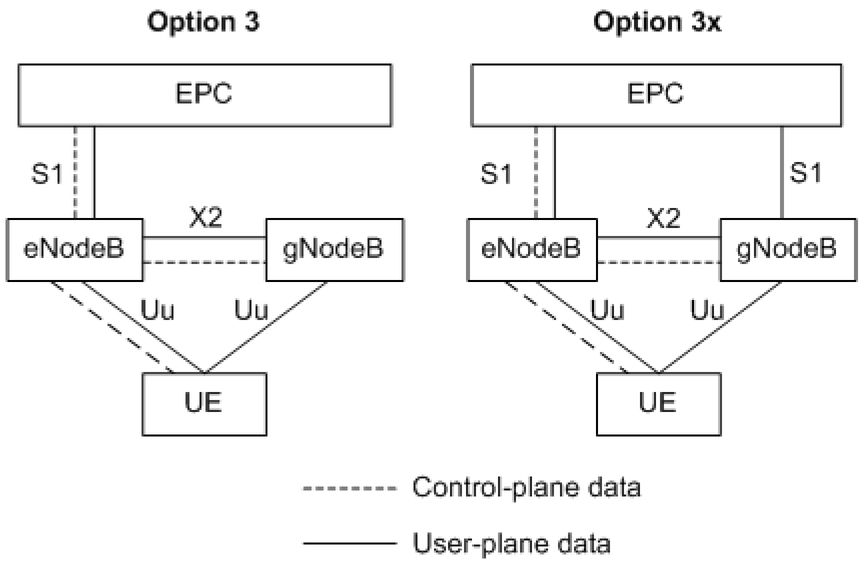
In NSA networking, the QoS management architecture of the gNodeB is the same as that of the eNodeB. EPS bearer is the finest granularity of QoS management. For details about the eNodeB QoS management architecture, see *QoS Management* in eRAN feature documentation.

In NSA networking, the EPC is still the core network. The eNodeB functions as the master eNodeB (MeNB) to process control-plane data for bearer setup, and the gNodeB functions as the secondary gNodeB (SgNB) for user-plane data processing. Therefore, the gNodeB QoS configuration information is derived from an SGNB ADDITION REQUEST or SGNB MODIFICATION REQUEST message over the X2 interface. The messages contain information required for bearer setup on the SgNB, such as the QoS class identifier (QCI), allocation and retention priority (ARP), maximum bit rate (MBR), guaranteed bit rate (GBR), and per UE aggregate maximum bit rate (UE-AMBR). Based on such information, the gNodeB provides differentiated QoS guarantee.

In NSA networking, Huawei supports Option 3 and Option 3x shown in **Figure 3-1**.

- With Option 3, the eNodeB is the data split anchor. The eNodeB can carry all user-plane data itself using the master cell group (MCG) bearer. The eNodeB can also distribute a portion of user-plane data to the gNodeB, using the MCG split bearer.
- With Option 3x, the gNodeB is the data split anchor. The gNodeB can distribute a portion of user-plane data to the eNodeB, using the secondary cell group (SCG) split bearer.

Figure 3-1 Option 3 and Option 3x



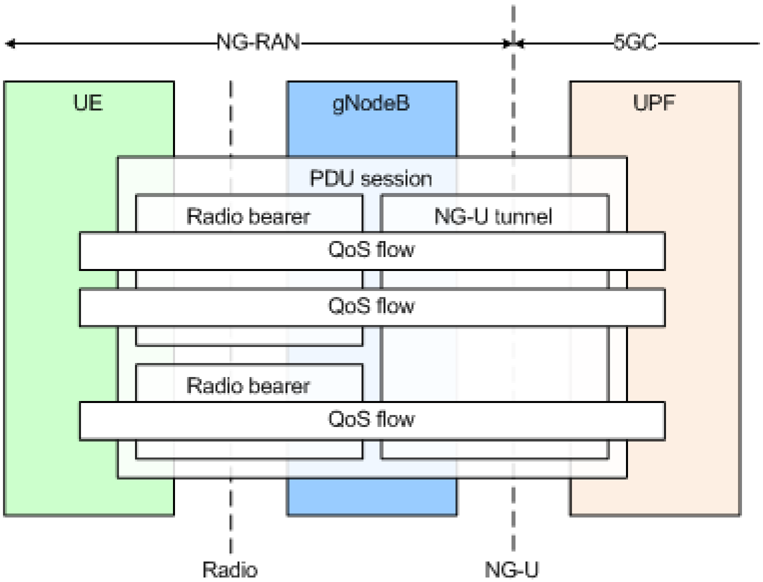
SA Networking

In SA networking, the 5G Core Network (5GC) serves as the core network and the QoS architecture has changed significantly, as shown in **Figure 3-2**.

The radio bearer is still used between the gNodeB and the UE. However, the NG-U tunnel, instead of the S1 bearer is used between the gNodeB and the core network, and the finest granularity of QoS management is changed from EPS bearer to QoS flow. Each QoS flow is identified by a QoS flow ID (QFI). In a PDU session, the QFI of each QoS flow is unique. Each QoS flow can carry only one service type. The service type's corresponding QoS level is identified by a unique 5G QoS Identifier (5QI). However, a service type with a QoS level identified by one 5QI can be carried by multiple QoS flows. The core network informs the gNodeB of the 5QI mapping each QoS flow to define the QoS attributes.

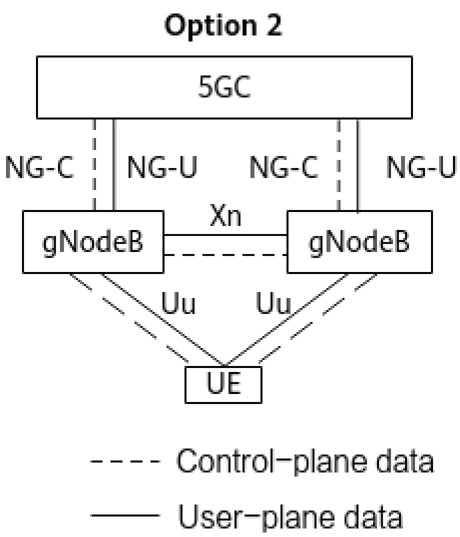
When a UE initiates a service, the 5GC maps the service's data flow to one or more QoS flows. Then, the gNodeB maps the QoS flows to data radio bearers (DRBs) over the air interface. There can be many-to-one or one-to-one mapping relationships between QoS flows and DRBs. When mapping QoS flows to DRBs, Huawei gNodeBs use QCIs to identify QoS levels of the DRBs. This is to ensure QoS architecture compatibility between NSA and SA networking.

Figure 3-2 QoS architecture in SA networking



In SA networking, Huawei supports Option 2 shown in **Figure 3-3**.

Figure 3-3 Option 2



4 QoS Management

4.1 Principles

4.1.1 QoS Management in the Service Setup Request Phase

In the service setup request phase, services are mapped to the QCI-specific bearers of the gNodeB in both NSA and SA networking. Differentiated service (DiffServ) is implemented by allocating different bearers.

4.1.1.1 NSA Service-to-Bearer Mapping

4.1.1.1.1 Overview

When a UE initiates a service setup request, the gNodeB allocates appropriate radio bearers and transmission resources for the service based on the QoS attributes of the service-specific QCI obtained over the X2 interface and the configurations on the gNodeB side.

4.1.1.1.2 Configuring QCIs and Corresponding QoS Attributes

Table 4-1 and **Table 4-2** provide the QoS attributes that map each standardized QCI in NSA networking for resource types GBR and non-GBR, respectively. For details, see section 6.1.7 "Standardized QoS characteristics" in 3GPP TS 23.203 V15.4.0.

QCIs and the corresponding QoS attributes (**Priority Level** and **Packet Delay Budget**) can be configured using the **gNBQciBearer** MO, in which:

- **gNBQciBearer**.***Qci*** is used to configure the QCI.
- **gNBQciBearer**.***PriorityLevel*** is used to configure **Priority Level**.

NOTE

For UEs in NSA networking, the default value of this parameter is 10 times the value of **Priority Level** (as listed in **Table 4-1** and **Table 4-2**) defined in section 6.1.7 "Standardized QoS characteristics" in 3GPP TS 23.203 V15.4.0.

- **gNBQciBearer**.***PacketDelayBudget*** is used to configure **Packet Delay Budget** (or referred to as PDB).

Table 4-1 Standardized QCI characteristics (GBR)

QCI	Priority Level	Packet Delay Budget	Packet Error Loss Rate	Example Services
1	2	100 ms	10^{-2}	Conversational voice
2	4	150 ms	10^{-3}	Conversational video (live streaming)
3	3	50 ms	10^{-3}	Real-time gaming, V2X message, power distribution (medium voltage), process automation (monitoring)
4	5	300 ms	10^{-6}	Non-conversational video (buffered streaming)
65	0.7	75 ms	10^{-2}	Mission-critical user-plane push-to-talk (PTT) voice, such as mission-critical push-to-talk (MCPTT)
66	2	100 ms	10^{-2}	Non-mission-critical user-plane PTT voice
67	1.5	100 ms	10^{-3}	Mission-critical user-plane video
75	2.5	50 ms	10^{-2}	V2X messages

Table 4-2 Standardized QCI characteristics (non-GBR)

QCI	Priority Level	Packet Delay Budget	Packet Error Loss Rate	Example Services
5	1	100 ms	10^{-6}	IP multimedia subsystem (IMS) signaling
6	6	300 ms	10^{-6}	Video (buffered streaming) TCP-based services (such as web browsing, email, chatting, FTP, P2P file sharing, and progressive video)

QCI	Priority Level	Packet Delay Budget	Packet Error Loss Rate	Example Services
7	7	100 ms	10^{-3}	Voice Video (live streaming) Interactive gaming
8	8	300 ms	10^{-6}	Video (buffered streaming) TCP-based services (such as web browsing, email, chatting, FTP, P2P file sharing, and progressive video)
9	9	300 ms	10^{-6}	
69	0.5	60 ms	10^{-6}	Mission-critical delay sensitive signaling (such as MCPTT signaling and mission-critical video signaling)
70	5.5	200 ms	10^{-6}	Mission-critical data (example services are the same as those with a QCI of 6/8/9)
79	6.5	50 ms	10^{-2}	V2X messages
80	6.8	10 ms	10^{-6}	Low latency eMBB applications (TCP/UDP-based) Augmented reality

4.1.1.1.3 Configuring Radio Bearers and Transmission Resource Parameters

Radio Bearer-related Parameters

Radio bearer-related parameters include those in PDCP, MAC, and RLC parameter groups.

- Parameters in a PDCP parameter group (base-station-level parameters of which are configurable using the **gNBPdcpParamGroup** MO)
  - The **NRCellQciBearer** MO on the gNodeB is used to configure the QCI-specific bearer of an NR cell. Settings of base-station-level parameters in PDCP parameter groups can be modified for bearers with different QCIs in different cells so that the cell-level parameter configurations take effect in each cell.
    - If the RLC mode for a bearer is unacknowledged mode (UM), parameters in the PDCP parameter group in UM corresponding to **NRCellQciBearer**.***UmPdcpParamGroupId*** take effect.

    - If the RLC mode for a bearer is acknowledged mode (AM), parameters in the PDCP parameter group in AM corresponding to **NRCellQciBearer**.***AmPdcpParamGroupId*** take effect.
  - The **NRCellOpQciBearer** MO on the gNodeB is used to configure the QCI-specific bearer of an operator in an NR cell. For bearers with the same QCI, the base-station-level PDCP parameter groups can be modified on a per operator basis so that the operator-level parameter configurations take effect for a cell.
    - If the RLC mode for a bearer is UM, parameters in the PDCP parameter group in UM corresponding to **NRCellOpQciBearer**.***UmPdcpParamGroupId*** take effect.
    - If the RLC mode for a bearer is AM, parameters in the PDCP parameter group in AM corresponding to **NRCellOpQciBearer**.***AmPdcpParamGroupId*** take effect.

  If different PDCP parameter groups are configured in the **NRCellQciBearer** and **NRCellOpQciBearer** MOs for the same QCI in a cell, the configurations in the **NRCellOpQciBearer** MO take effect.

- Parameters in a MAC parameter group (base-station-level parameters of which are configurable using the **gNBDUMacParamGroup** MO)

  The **NRDUCellQciBearer** MO on the gNodeB can be used to configure the QCI-specific bearer of an NR DU cell. Settings of base-station-level parameters in MAC parameter groups can be modified for bearers with different QCIs in different cells so that the cell-level parameter configurations (corresponding to **NRDUCellQciBearer.*MacParamGroupId***) take effect in each cell.

- Parameters in an RLC parameter group (base-station-level parameters of which are configurable using the **gNBRlcParamGroup** MO)

  The **NRDUCellQciBearer** MO on the gNodeB can be used to configure the QCI-specific bearer of an NR DU cell. Settings of base-station-level parameters in RLC parameter groups can be modified for bearers with different QCIs in different cells so that the cell-level parameter configurations take effect in each cell.

  - If the RLC mode for a bearer is UM, parameters in the RLC parameter group in UM corresponding to **NRDUCellQciBearer**.***UmRLCParamGroupId*** take effect.
  - If the RLC mode for a bearer is AM, parameters in the RLC parameter group in AM corresponding to **NRDUCellQciBearer**.***AmRLCParamGroupId*** take effect.

NOTE

The RLC mode of a bearer, specified by the **NRCellQciBearer**.***RlcMode*** or **NRCellOpQciBearer**.***RlcMode*** parameter, can be set to **UM** or **AM**.

- If the SGNB ADDITION REQUEST or SGNB MODIFICATION REQUEST message over the X2 interface carries the *RLC Mode* IE, the configuration of this IE is used.
- If the SGNB ADDITION REQUEST or SGNB MODIFICATION REQUEST message over the X2 interface does not carry the *RLC Mode* IE, the RLC mode specified by the **NRCellQciBearer**.***RlcMode*** or **NRCellOpQciBearer**.***RlcMode*** parameter is used. If different RLC modes are configured in the **NRCellQciBearer** and **NRCellOpQciBearer** MOs for the same QCI in a cell, the configurations in the **NRCellOpQciBearer** MO take effect.

If the SGNB ADDITION REQUEST message over the X2 interface requires a bearer with a new QCI to be set up:

- If this QCI is not configured in both the **NRCellQciBearer** and **NRCellOpQciBearer** MOs, parameter configurations in the PDCP parameter group corresponding to this new QCI take effect with the **NRCellQciBearer**.***Qci*** parameter value being **9**.
- If this QCI is configured in the **NRCellOpQciBearer** MO, parameter configurations in the PDCP parameter group corresponding to this new QCI take effect based on the value of the **NRCellOpQciBearer**.***Qci*** parameter.
- If this QCI is configured in the **NRCellQciBearer** MO, but not in the **NRCellOpQciBearer** MO, parameter configurations in the PDCP parameter group corresponding to this new QCI take effect based on the value of the **NRCellQciBearer**.***Qci*** parameter.
- If this QCI is not configured in the **NRDUCellQciBearer** MO, parameter configurations in the MAC and RLC parameter groups corresponding to this new QCI take effect with the **NRDUCellQciBearer**.***Qci*** parameter value being **9**.
- If this QCI is configured in the **NRDUCellQciBearer** MO, parameter configurations in the MAC parameter group and RLC parameter group corresponding to this new QCI take effect based on the value of the **NRDUCellQciBearer**.***Qci*** parameter.

Transmission Resource-related Parameters

The S1 and X2 interfaces use the IP transmission technology. DiffServ is introduced to ensure QoS for IP transmission. The differentiated services code point (DSCP) field is contained in the header of IP packets to signal the QoS requirements to each router on a transmission path. The DSCP value ranges from 0 to 63, with a larger value indicating a higher scheduling priority. For details about the transmission resource-related parameter configuration, see *Transmission Resource Management*.

4.1.1.2 SA Service-to-Bearer Mapping

4.1.1.2.1 Overview

When UEs initiate service setup requests, the gNodeB reads the QoS attributes of each QoS flow in INITIAL CONTEXT SETUP REQUEST or PDU SESSION RESOURCE SETUP REQUEST messages over the NG interface. The gNodeB maps QoS flows to the corresponding DRBs based on the 5QI of each QoS flow and the configurations on the gNodeB side, and allocates appropriate radio bearers and transmission resources to services.

4.1.1.2.2 Configuring 5QIs and Corresponding QoS Attributes

Table 4-3, **Table 4-4**, and **Table 4-5** provide the QoS attributes that map each standardized 5QI in SA networking for resource types GBR, non-GBR, and delay-critical GBR, respectively. For details, see section 5.7.4 "Standardized 5QI to QoS characteristics mapping" in 3GPP TS 23.501 V16.6.0.

The 5QI is delivered by the core network to the base station. On the base station side, the **gNB5qiConfig**.***Nr5qi*** parameter specifies the 5QI, and the

gNB5qiConfig.***Qci*** parameter specifies the QCI corresponding to the 5QI. QCIs and the corresponding QoS attributes (**Default Priority Level**, **Packet Delay Budget**, and **Default Averaging Window**) can be configured using the **gNBQciBearer** MO, in which:

- **gNBQciBearer**.***Qci*** is used to configure the QCI.
- **gNBQciBearer**.***PriorityLevel*** is used to configure **Default Priority Level**.

  NOTE

  - **gNBQciBearer**.***PriorityLevel*** is used on Huawei base stations to configure **Priority Level** to ensure consistency between NSA and SA networking.
  - For UEs in SA networking, the default value of this parameter is the same as the value of **Priority Level** (as listed in **Table 4-3**, **Table 4-4**, and **Table 4-5**) defined in section 5.7.4 "Standardized 5QI to QoS characteristics mapping" in 3GPP TS 23.501 V15.4.0.
- **gNBQciBearer**.***PacketDelayBudget*** is used to configure **Packet Delay Budget**.

  If the base station receives an extended PDB (configured on the core network), the base station preferentially uses this extended PDB configuration. For details, see section 9.3.1.18 "Dynamic 5QI Descriptor" in 3GPP TS 38.413 V17.6.0.
- **gNBQciBearer**.***AveragingWindowDur*** is used to configure **Default Averaging Window**.

Table 4-3 Standardized 5QI to QoS characteristics mapping (GBR)

5QI	Default Priority Level	Packet Delay Budget	Packet Error Rate	Default Maximum Data Burst Volume	Default Averaging Window	Example Services
1	20	100 ms	10^{-2}	N/A	2000 ms	Conversational voice
2	40	150 ms	10^{-3}	N/A	2000 ms	Conversational video (live streaming)
3	30	50 ms	10^{-3}	N/A	2000 ms	Real-time gaming, V2X messages Power distribution (medium voltage), process automation (monitoring)
4	50	300 ms	10^{-6}	N/A	2000 ms	Non-conversational video (buffered streaming)
65	7	75 ms	10^{-2}	N/A	2000 ms	Mission-critical user-plane PTT voice, such as MCPTT
66	20	100 ms	10^{-2}	N/A	2000 ms	Non-mission-critical user-plane PTT voice

5QI	Default Priority Level	Packet Delay Budget	Packet Error Rate	Default Maximum Data Burst Volume	Default Averaging Window	Example Services
67	15	100 ms	10^{-3}	N/A	2000 ms	Mission-critical user-plane video
71	56	150 ms	10^{-6}	N/A	2000 ms	Uplink live streaming
72	56	300 ms	10^{-4}	N/A	2000 ms	Uplink live streaming
73	56	300 ms	10^{-8}	N/A	2000 ms	Uplink live streaming
74	56	500 ms	10^{-8}	N/A	2000 ms	Uplink live streaming
76	56	500 ms	10^{-4}	N/A	2000 ms	Uplink live streaming

Table 4-4 Standardized 5QI to QoS characteristics mapping (non-GBR)

5QI	Default Priority Level	Packet Delay Budget	Packet Error Rate	Default Maximum Data Burst Volume	Default Averaging Window	Example Services
5	10	100 ms	10^{-6}	N/A	N/A	IMS signaling
6	60	300 ms	10^{-6}	N/A	N/A	Video (buffered streaming) TCP-based services (such as web browsing, email, chatting, FTP, P2P file sharing, and progressive video)
7	70	100 ms	10^{-3}	N/A	N/A	Voice Video (live streaming) Interactive gaming
8	80	300 ms	10^{-6}	N/A	N/A	Video (buffered streaming) TCP-based services (such as web browsing, email, chatting, FTP, P2P file sharing, and progressive video)
9	90	300 ms	10^{-6}	N/A	N/A	

5QI	Default Priority Level	Packet Delay Budget	Packet Error Rate	Default Maximum Data Burst Volume	Default Averaging Window	Example Services
69	5	60 ms	10^{-6}	N/A	N/A	Mission-critical delay sensitive signaling (such as MCPTT signaling)
70	55	200 ms	10^{-6}	N/A	N/A	Mission-critical data (example services are the same as those with a QCI of 6/8/9)
79	65	50 ms	10^{-2}	N/A	N/A	V2X messages
80	68	10 ms	10^{-6}	N/A	N/A	Low latency eMBB application Augmented reality

Table 4-5 Standardized 5QI to QoS characteristics mapping (delay-critical GBR)

5QI	Default Priority Level	Packet Delay Budget	Packet Error Rate	Default Maximum Data Burst Volume	Default Averaging Window	Example Services
82	19	10 ms	10^{-4}	255 bytes	2000 ms	Discrete automation
83	22	10 ms	10^{-4}	1354 bytes	2000 ms	Discrete automation and V2X messages (vehicle platooning and advanced driving, such as collaborative lane changing with a low level of automation (LoA))
84	24	30 ms	10^{-5}	1354 bytes	2000 ms	Intelligent transportation systems
85	21	5 ms	10^{-5}	255 bytes	2000 ms	Electricity distribution (high voltage) and V2X messages (remote driving)

5QI	Default Priority Level	Packet Delay Budget	Packet Error Rate	Default Maximum Data Burst Volume	Default Averaging Window	Example Services
86	18	5 ms	10^{-4}	1354 bytes	2000 ms	V2X messages (advanced driving, such as vehicle platooning with a high LoA and collision avoidance)

The 5QI can either be a dynamic 5QI or a non-dynamic 5QI, which is assigned by the core network.

The core network delivers QoS attribute parameters for dynamic 5QIs to the gNodeB, whereas those for non-dynamic 5QIs are optional parameters that the core network may deliver or not.

- If the core network delivers the QoS attribute parameters, the values of the delivered parameters take effect. The gNodeB reads the INITIAL CONTEXT SETUP REQUEST or PDU SESSION RESOURCE SETUP REQUEST message over the NG interface to obtain the QoS attributes of each QoS flow.
- If the core network does not deliver QoS attribute parameters, the QoS attributes that take effect depend on the configurations of the **gNB5qiConfig** and **gNBQciBearer** MOs on the gNodeB side.

  The **gNB5qiConfig** MO configuration is checked to query the QCI (**gNB5qiConfig**.***Qci***) corresponding to a 5QI (**gNB5qiConfig**.***Nr5qi***) assigned by the core network.

  – When the QCI corresponding to the 5QI assigned by the core network is successfully queried, a further check is made on whether this QCI has been configured (using **gNBQciBearer**.***Qci***) in the **gNBQciBearer** MO.

    ▪ If so, the QoS attributes corresponding to the QCI configured in the **gNBQciBearer** MO take effect.

    ▪ If not, the QoS attributes corresponding to the **gNB5qiConfig**.***Qci*** parameter value being **9** take effect.

  – When there is no QCI corresponding to the 5QI assigned by the core network, the QoS attributes corresponding to the **gNB5qiConfig**.***Nr5qi*** parameter value being **9** take effect.

Compatibility Processing

GBR Service Type Consistency Check

To check the consistency between the 5QI-specific service type (GBR) of a QoS flow and the core-network-delivered one, the **GBR_ADMIT_BY_23501_SW** option of the **gNodeBParam**.***CompatibilityAlgoSwitch*** parameter can be used to specify whether to enable the consistency check function.

- When this function is enabled:
  – If they are consistent, the QoS flow is established successfully.
  – If they are inconsistent, the QoS flow establishment fails.
- When this function is disabled:
  – If they are consistent, the QoS flow is established successfully.
  – If they are inconsistent, the core-network-delivered service type is used, and the QoS flow can still be established successfully.

NOTE

The core-network-delivered service type of a QoS flow is GBR if *GBR QoS Flow Information* is contained in the *QoS Flow Level QoS Parameters* IE in the PDU Session Resource Modify message delivered by the core network. Otherwise, the service type is non-GBR.

PDU Session Resource Modify Compatibility Optimization

The **PDU_SESSION_MOD_COMPAT_OPT_SW** option of the **gNodeBParam**.***CompatibilityAlgoSwitch*** parameter specifies whether to enable the PDU Session Resource Modify compatibility optimization function.

When the PDU Session Resource Modify compatibility optimization function is enabled:

- The base station can change GBR services of the same QoS flow to non-GBR services and change non-GBR services of the same QoS flow to GBR services.
- The base station can change dynamic 5QIs to non-dynamic 5QIs, in which case the core-network-delivered 5QIs are used. The base station can also change non-dynamic 5QIs to dynamic 5QIs, in which case the core-network-delivered 5QIs are used or the 5QI of 9 is used when the core network does not deliver 5QIs.
- In the event of QoS flow modification, the QoS flow attribute is determined by the *QoS Flow Add or Modify Request Item* IE carried in the PDU Session Resource Modify message:
  – If the core network delivers this IE, the processing by the base station complies with the attribute indicated by this IE.
  – If the core network does not deliver this IE, the configured value is used.

When the PDU Session Resource Modify compatibility optimization function is disabled:

- The base station cannot change GBR services of the same QoS flow to non-GBR services or change non-GBR services of the same QoS flow to GBR services. In this case, the base station includes the *PDU Session Resource Modify Unsuccessful Transfer* IE in the PDU SESSION RESOURCE MODIFY RESPONSE message.
- The base station cannot change dynamic 5QIs to non-dynamic 5QIs or change non-dynamic 5QIs to dynamic 5QIs. In this case, the base station includes the *PDU Session Resource Modify Unsuccessful Transfer* IE in the PDU SESSION RESOURCE MODIFY RESPONSE message.

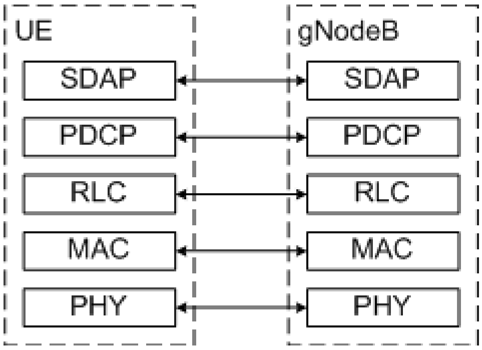
PDU Session Resource Modify QCI Mapping Optimization

When the PDU Session Resource Modify compatibility optimization function is enabled, the gNodeB re-maps 5QIs to QCIs even if the 5QI values do not change after the core network delivers the *QoS Flow Add or Modify Request Item* IE to modify the QoS flow attributes. The remapping may interrupt some assurance features. Against this backdrop, the PDU Session Resource Modify QCI mapping optimization function (controlled by the **PDU_SESSION_MOD_QCI_MAP_OPT_SW** option of the **gNodeBParam**.***CompatibilityAlgoSwitch*** parameter) is introduced. After this function is enabled and the *QoS Flow Add or Modify Request Item* IE is modified for a QoS flow, QCI mapping is re-performed only if the 5QI value changes, thereby avoiding service interruption caused by remapping.

4.1.1.2.3 QoS Flow-to-DRB Mapping

Service Data Adaptation Protocol (SDAP) is a protocol layer added in 5G. **Figure 4-1** shows how this protocol layer fits in the 5G user-plane protocol stack.

Figure 4-1 5G user-plane protocol stack



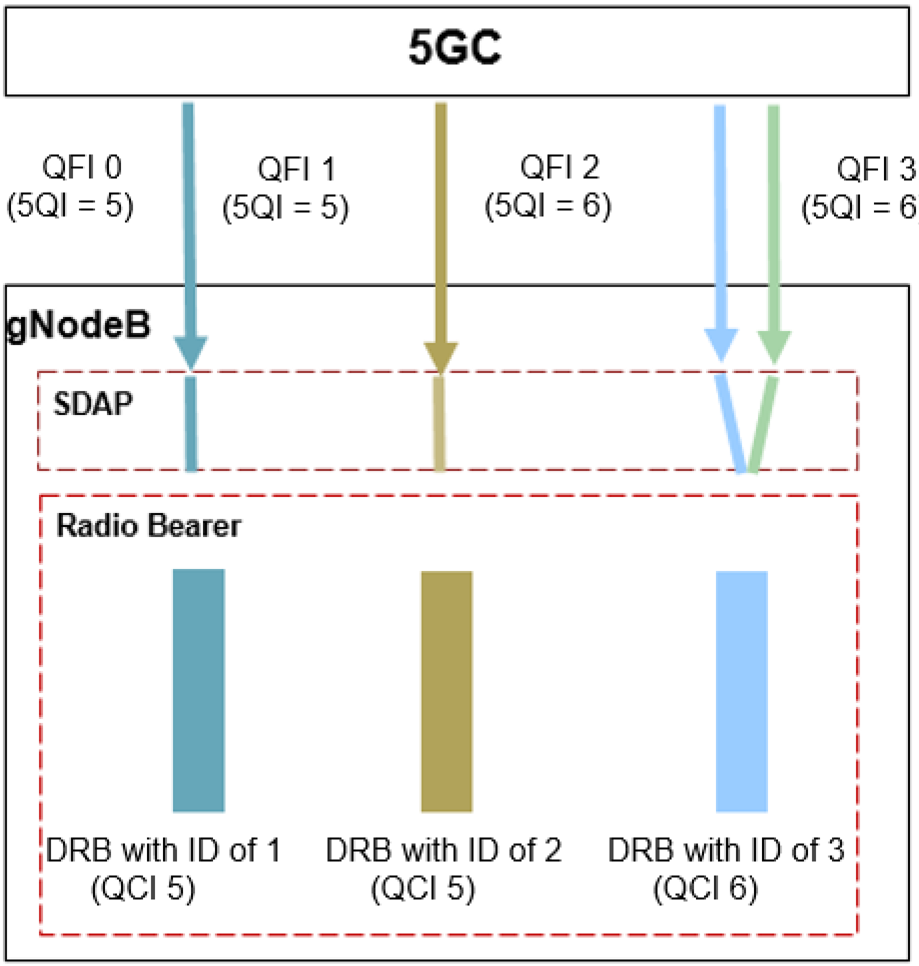
The SDAP layer provides two functions:

- Adds the QFI in a data packet. The receiver reads the QFI from the SDAP header of the data packet.
- Maps QoS flows onto corresponding DRBs.

When QoS flows pass through the SDAP layer, the SDAP layer maps the QoS flows to the corresponding DRBs.

- If the QoS flows correspond to GBR or non-GBR services with a 5QI of 5 or 69, the gNodeB maps each QoS flow to a DRB separately. In this scenario, QoS flows and DRBs are in one-to-one mapping relationships. As shown in **Figure 4-2**, QFI 0 and QFI 1 with the 5QI of 5 (mapping two QoS flows of the same service type) are mapped to the DRBs with the IDs of 1 and 2, respectively. However, the QCI of both the DRBs is 5.
- If the QoS flows correspond to the non-GBR service type with a 5QI other than 5 or 69:
  - When the *sdap-HeaderDL* or *sdap-HeaderUL* IE in RRC signaling is set to **absent**, the gNodeB maps each QoS flow to a DRB separately. In this scenario, QoS flows and DRBs are in one-to-one mapping relationships, as shown in **Figure 4-2**.
  - When the *sdap-HeaderDL* and *sdap-HeaderUL* IEs in RRC signaling are both set to **present**, the gNodeB maps multiple QoS flows of the same service type onto the same DRB. In this scenario, QoS flows and DRBs are in many-to-one mapping relationships. As shown in **Figure 4-2**, QFI 2 and QFI 3 with the 5QI of 6 (mapping two QoS flows of the same service type) are mapped to the DRB with the ID of 3.

Figure 4-2 QoS flow-to-DRB mapping



When a 5QI listed in **Table 4-6** is added by the core network, its mapping with a QCI is created by the gNodeB by default. The QCIs mapped to 5QI 1 to 5QI 9, 5QI 65, 5QI 66, 5QI 69, 5QI 70, 5QI 75, and 5QI 79 cannot be modified.

When a 5QI that is not listed in **Table 4-6** is added by the core network, its mapping with a QCI must be manually configured. For detailed configurations, see **4.4.1.2 Using MML Commands**.

Table 4-6 5QI-QCI mapping created by the gNodeB by default

5QI	Mapped QCI
1	1
2	2
3	3
4	4
5	5
6	6

5QI	Mapped QCI
7	7
8	8
9	9
65	65
66	66
67	67
69	69
70	70
71	71
72	72
73	73
74	74
75	75
76	76
79	79
80	80
82	82
83	83
84	84
85	85
86	86

NOTE

5QI 75 is currently not supported in 3GPP TS 23.501 V16.5.0.

When the core network delivers a 5QI to the gNodeB, the 5QI-QCI mapping rules are as follows:

- If the 5QI has been configured in the **gNB5qiConfig** MO, a further check is made on whether its corresponding QCI (**gNB5qiConfig**.***Qci***) has been configured using the NR cell-specific parameter **NRCellQciBearer**.***Qci*** or **NRCellOpQciBearer**.***Qci*** and the NR DU cell-specific parameter **NRDUCellQciBearer**.***Qci***.
  - If so, the configurations corresponding to this QCI take effect.

  - If not, the configurations corresponding to the **NRCellQciBearer**.***Qci*** and **NRDUCellQciBearer**.***Qci*** parameter values being **9** take effect by default.
- If this 5QI has not been configured in the **gNB5qiConfig** MO, the configurations corresponding to the **gNB5qiConfig**.***Nr5qi*** parameter value being **9** take effect by default.

In the downlink, the gNodeB maps QoS flows to DRBs so that data can be sent to the UEs over the mapped bearers.

In the uplink, UEs map QoS flows to DRBs in either of the following modes so that data can be sent to the gNodeB over the mapped bearers. UEs use the latest mapping between QoS flows and DRBs in either mode.

- Reflecting mapping mode

  If the **gNB5qiConfig**.***AsReflectiveQosSwitch*** parameter is set to **ON**, when the gNodeB receives a QoS Flows Add Request or QoS Flows Modify Request message over the NG interface:
  - If the gNodeB does not need to send an RRCReconfiguration message to deliver QoS flow-to-DRB mapping information (for example, because the current DRB meets the QoS flow requirements), reflecting mapping can be used by UEs to find the DRBs corresponding to data packets to reduce signaling overheads. Specifically, a UE monitors the QFI marked in each downlink data packet on each DRB, records the mapping between QFI-specific downlink data packets and DRBs, and maps the uplink data packets with the same QFI to the same DRB. For details about this mode, see chapter 12 "QoS" in 3GPP TS 38.300 V15.4.0.
  - If the gNodeB needs to deliver QoS flow-to-DRB mapping in an RRCReconfiguration message (for example, because the existing DRB does not meet the QoS flow requirements or due to other reasons), the RRCReconfiguration message is sent to the UE.
- Explicit reconfiguration mode

  If the **gNB5qiConfig**.***AsReflectiveQosSwitch*** parameter is set to **OFF**, the gNodeB sends an RRCReconfiguration message containing the mapping between QoS flows and DRBs to UEs.

4.1.1.2.4 Configuring Radio Bearers and Transmission Resource Parameters

Radio Bearer-related Parameters

Radio bearer-related parameters include those in PDCP, MAC, and RLC parameter groups.

- Parameters in a PDCP parameter group (base-station-level parameters of which are configurable using the **gNBPdcpParamGroup** MO)
  - The **NRCellQciBearer** MO on the gNodeB is used to configure the QCI-specific bearer of an NR cell. Settings of base-station-level parameters in PDCP parameter groups can be modified for bearers with different QCIs in different cells so that the cell-level parameter configurations take effect in each cell.
    - If the RLC mode for a bearer is unacknowledged mode (UM), parameters in the PDCP parameter group in UM corresponding to **NRCellQciBearer**.***UmPdcpParamGroupId*** take effect.

    - If the RLC mode for a bearer is acknowledged mode (AM), parameters in the PDCP parameter group in AM corresponding to **NRCellQciBearer**.***AmPdcpParamGroupId*** take effect.

  – The **NRCellOpQciBearer** MO on the gNodeB is used to configure the QCI-specific bearer of an operator in an NR cell. For bearers with the same QCI, the base-station-level PDCP parameter groups can be modified on a per operator basis so that the operator-level parameter configurations take effect for a cell.

    - If the RLC mode for a bearer is UM, parameters in the PDCP parameter group in UM corresponding to **NRCellOpQciBearer**.***UmPdcpParamGroupId*** take effect.

    - If the RLC mode for a bearer is AM, parameters in the PDCP parameter group in AM corresponding to **NRCellOpQciBearer**.***AmPdcpParamGroupId*** take effect.

  If different PDCP parameter groups are configured in the **NRCellQciBearer** and **NRCellOpQciBearer** MOs for the same QCI in a cell, the configurations in the **NRCellOpQciBearer** MO take effect.

- Parameters in a MAC parameter group (base-station-level parameters of which are configurable using the **gNBDUMacParamGroup** MO)

  The **NRDUCellQciBearer** MO on the gNodeB can be used to configure the QCI-specific bearer of an NR DU cell. Settings of base-station-level parameters in MAC parameter groups can be modified for bearers with different QCIs in different cells so that the cell-level parameter configurations (corresponding to **NRDUCellQciBearer.*MacParamGroupId***) take effect in each cell.

- Parameters in an RLC parameter group (base-station-level parameters of which are configurable using the **gNBRlcParamGroup** MO)

  The **NRDUCellQciBearer** MO on the gNodeB can be used to configure the QCI-specific bearer of an NR DU cell. Settings of base-station-level parameters in RLC parameter groups can be modified for bearers with different QCIs in different cells so that the cell-level parameter configurations take effect in each cell.

  – If the RLC mode for a bearer is UM, parameters in the RLC parameter group in UM corresponding to **NRDUCellQciBearer**.***UmRLCParamGroupId*** take effect.

  – If the RLC mode for a bearer is AM, parameters in the RLC parameter group in AM corresponding to **NRDUCellQciBearer**.***AmRLCParamGroupId*** take effect.

The RLC mode of a bearer, specified by the **NRCellQciBearer**.***RlcMode*** or **NRCellOpQciBearer**.***RlcMode*** parameter, can be set to **UM** or **AM**. If different RLC modes are configured in the **NRCellQciBearer** and **NRCellOpQciBearer** MOs for the same QCI in a cell, the configurations in the **NRCellOpQciBearer** MO take effect.

- For bearers with a QCI of 4, 5, 6, 8, 9, 69, 70, 71, 72, 73, 74, 76, or 80, a newly deployed gNodeB automatically uses **AM** as the value of **NRCellQciBearer**.***RlcMode***.

- For bearers with a QCI of 1, 2, 3, 7, 65, 66, 67, 75, 79, 82, 83, 84, 85, or 86, a newly deployed gNodeB automatically uses **UM** as the value of **NRCellQciBearer**.***RlcMode***.
- After being automatically configured on the gNodeB based on the QCI, the **NRCellQciBearer**.***RlcMode*** parameter can be manually reconfigured according to the network plan. However, it is recommended that the **NRCellQciBearer**.***RlcMode*** parameter be set to **UM** for bearers with a QCI of 1, 65, or 66.

Transmission Resource-related Parameters

The N3 and Xn interfaces use the IP transmission technology. DiffServ is introduced to ensure QoS for IP transmission. The differentiated services code point (DSCP) field is contained in the header of IP packets to signal the QoS requirements to each router on a transmission path. The DSCP value ranges from 0 to 63, with a larger value indicating a higher scheduling priority. For details about the transmission resource-related parameter configuration, see *Transmission Resource Management*.

4.1.2 QoS Management for Established Services

QoS management for established services can be classified into the following types based on current procedures:

- QoS guarantee for scheduling procedure: It is categorized into QoS guarantee for downlink scheduling and QoS guarantee for uplink scheduling. QoS guarantee for scheduling procedure requires the base station to calculate the scheduling priorities of bearers based on the QoS attributes (such as QCI/5QI and **Packet Delay Budget**), channel quality, rate requirements, and bearer weights to maximize the system traffic volume while fulfilling QoS requirements of UEs. For more information about QoS guarantee for downlink scheduling and uplink scheduling after service establishment, see **4.1.2.1 QoS Guarantee for Downlink Scheduling** and **4.1.2.2 QoS Guarantee for Uplink Scheduling**, respectively.
- QoS guarantee for non-scheduling procedure: For details, see **4.1.2.3 QoS Guarantee for Non-Scheduling Procedure**.

4.1.2.1 QoS Guarantee for Downlink Scheduling

The gNodeB alone can implement QoS guarantee for downlink scheduling. It can obtain the data volume of each downlink bearer and based on inputs such as the QoS parameters, channel quality, and historical rate, determine the scheduling priorities of bearers and select the bearer to be scheduled. The downlink basic scheduling priority is calculated using the following formula:

$$\text{priority} = \frac{\text{eff}}{r} \times \gamma_{\text{conn}}$$

where

- eff: Indicates the channel quality of a UE.
- r: Indicates the historical transmission rate of a UE.

- Y_{conn}: Indicates the downlink scheduling priority weighting factor of a specific service type. A larger factor indicates a higher downlink scheduling priority.
  - If the gNodeB does not obtain the RAT/Frequency Selection Priority (RFSP) index of a UE or the RFSP-relevant downlink scheduling priority coefficient (**gNBDURfspConfig**.***DlSchPriCoefficient***) is not configured for the UE: $Y_{conn} = Y(QCI)$.
  - If the gNodeB obtains the RFSP index of a UE and the RFSP-relevant downlink scheduling priority coefficient (**gNBDURfspConfig**.***DlSchPriCoefficient***) is configured for the UE: $Y_{conn} = Y(QCI) \times Coeff_{RFSP}$. In this case, RFSP-based downlink scheduling priority control applies.

  In the formula: $Y(QCI)$ indicates the QCI-specific bearer scheduling priority weighting factor (specified by **gNBDUMacParamGroup.*DlSchPriWeightFactor***) or QCI- and ARP-based bearer scheduling priority weighting factor (specified by **gNBArp**.***DlSchPriWeightFactor***). For details about the scheduling priority weighting for bearers with different QCIs or ARPs, see **4.1.2.1.5 ARP- and QCI-based Scheduling Priority Control**. $Coeff_{RFSP}$ indicates the downlink scheduling priority coefficient (specified by the **gNBDURfspConfig**.***DlSchPriCoefficient*** parameter) for UEs with the specified RFSP index. An operator can configure $Coeff_{RFSP}$ for specific UEs when desiring to customize downlink scheduling priorities for the UEs. For details about RFSP, see *Flexible User Steering*.

The scheduling type of a bearer is configured using the **gNBDUMacParamGroup**.***SchedulingType*** parameter.

QoS guarantee for downlink scheduling includes **4.1.2.1.1 UE-AMBR-based Rate Limitation for Non-GBR Services**, **4.1.2.1.2 Minimum Rate Guarantee for Non-GBR Services**, **4.1.2.1.3 Minimum Rate Guarantee for GBR Services**, **4.1.2.1.4 Maximum Rate Limitation for GBR Services**, **4.1.2.1.5 ARP- and QCI-based Scheduling Priority Control**, **4.1.2.1.6 Scheduling Optimization for GBR and Non-GBR Services**, and **4.1.2.1.7 Downlink PDB-based Differentiated Scheduling**.

4.1.2.1.1 UE-AMBR-based Rate Limitation for Non-GBR Services

Downlink UE-AMBR-based rate limitation for non-GBR services targets at all downlink non-GBR bearers of a UE. The **DL_AMBR_RATE_CTRL_SW** option of the **NRDUCellAlgoSwitch**.***ServiceDiffSwitch*** parameter can be selected to enable downlink UE-AMBR-based rate limitation for non-GBR services. After this function takes effect, the gNodeB limits the total rate of all downlink non-GBR bearers of a UE so that the total rate does not exceed the downlink UE-AMBR.

SA Networking

In SA networking, the downlink UE-AMBR value is min{Value of the *UE Aggregate Maximum Bit Rate Downlink* IE, Sum of the values of the *PDU Session Aggregate Maximum Bit Rate* IE of all the PDU sessions established by a UE}. The sources of the *UE Aggregate Maximum Bit Rate Downlink* and *PDU Session Aggregate Maximum Bit Rate* IEs are as follows:

- *UE Aggregate Maximum Bit Rate Downlink* IE: carried in an INITIAL CONTEXT SETUP REQUEST or PDU SESSION RESOURCE SETUP REQUEST message sent from the 5GC to the gNodeB over the NG interface
- *PDU Session Aggregate Maximum Bit Rate* IE: carried in *PDU Session Resource Setup Request Item* in a PDU SESSION RESOURCE SETUP REQUEST message or INITIAL CONTEXT SETUP REQUEST message sent from the 5GC to the gNodeB over the NG interface

The gNodeB performs downlink UE-AMBR-based rate control for non-GBR services. If there is data to be transmitted in the downlink buffer, a check is made in each TTI on whether the transmission traffic volume of all non-GBR bearers of a UE within a period of *T* is greater than the downlink UE-AMBR multiplied by *T*:

- If so, data split and scheduling of each bearer are suspended.
- If not, the priority calculated by basic scheduling is retained.

T is 3000 ms.

NSA Networking

In NSA networking, the downlink UE-AMBR value is indicated by the *SgNB UE Aggregate Maximum Bit Rate* IE in an SGNB ADDITION REQUEST message sent from the eNodeB to the gNodeB over the X2 interface.

In NSA DC scenarios, only the data split anchor can obtain the sum of the data volume carried on a bearer in both the MCG ($Volume_{MCG}$) and SCG ($Volume_{SCG}$). Therefore, rate control is only implemented by the base station that functions as the data split anchor. If there is data to be transmitted in the downlink buffer, a check is made in each TTI on whether the transmission traffic volume of all non-GBR bearers of a UE within a period of *T* is greater than the downlink UE-AMBR multiplied by *T*:

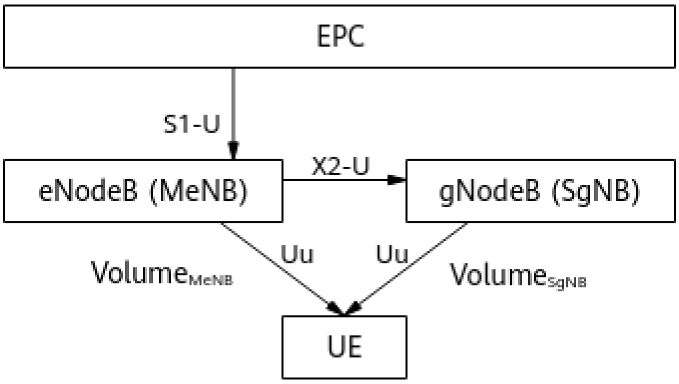
- If so, data split and scheduling of each bearer are suspended.
- If not, the priority calculated by basic scheduling is retained.

T is 1000 ms. The following describes the processing modes with Option 3 and Option 3x in NSA networking.

Option 3 (eNodeB as the Data Split Anchor)

As shown in **Figure 4-3**, the eNodeB performs UE-AMBR-based rate limitation, whereas the gNodeB only needs to perform such rate limitation on traffic distributed to it. Based on the value of the *UE Aggregate Maximum Bit Rate Downlink* IE, the gNodeB performs rate limitation on the total data ($Volume_{SgNB}$) transmitted on all non-GBR bearers of a UE.

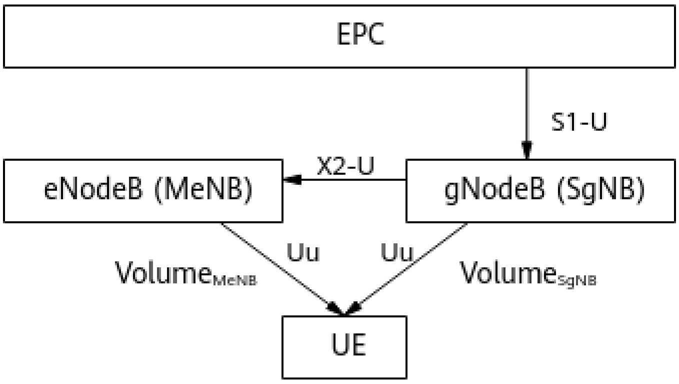
Figure 4-3 Downlink data split with Option 3



Option 3x (gNodeB as the Data Split Anchor)

As shown in **Figure 4-4**, the gNodeB performs UE-AMBR-based rate limitation, whereas the eNodeB only needs to perform such rate limitation on traffic distributed to it. Based on the value of the *UE Aggregate Maximum Bit Rate Downlink* IE, the gNodeB performs rate limitation on the total data ($Volume_{MeNB}$ plus $Volume_{SgNB}$) transmitted on all non-GBR bearers of a UE.

Figure 4-4 Downlink data split with Option 3x



4.1.2.1.2 Minimum Rate Guarantee for Non-GBR Services

Minimum rate guarantee for non-GBR services is used to ensure the minimum downlink rate for each non-GBR bearer.

If there is data to be transmitted in the downlink buffer, a check is made in each TTI on whether the transmission traffic volume of a non-GBR bearer within a period of *T* is less than the product of the **gNBDUMacParamGroup.*DlGuaranteedRate*** parameter value and *T*.

- If so, radio resources are preferentially allocated to the bearer. This ensures that its rate reaches the value of the **gNBDUMacParamGroup.*DlGuaranteedRate*** parameter.
- If not, the priority calculated by basic scheduling is retained.

In NSA networking, *T* is 1000 ms. In SA networking, *T* is 3000 ms.

NOTE

If the **gNBDUMacParamGroup.*DlGuaranteedRate*** parameter is set to **0**, rate guarantee for downlink non-GBR services is disabled. Generally, non-GBR services have a lower priority and a larger traffic volume than GBR services. Therefore, enabling rate guarantee for downlink non-GBR services may consume more resources. It is recommended that this function be used only when the minimum rate of downlink non-GBR services needs to be guaranteed.

In NSA DC scenarios, only the base station serving as the data split anchor can obtain the sum of the data volume carried on a bearer in both the MCG ($Volume_{MCG}$) and SCG ($Volume_{SCG}$). Therefore, rate control is only implemented by this base station. The following separately describes the processing in NSA networking with Option 3 and Option 3x.

With Option 3

With Option 3, the eNodeB guarantees the minimum rate for each non-GBR bearer of UEs. The gNodeB guarantees the minimum rate only for traffic distributed to it based on the **gNBDUMacParamGroup.*DlGuaranteedRate*** configuration. **Figure 4-3** shows downlink data split with Option 3.

With Option 3x

With Option 3x, the gNodeB guarantees the minimum rate for each non-GBR bearer of UEs. The eNodeB only guarantees the minimum rate for traffic distributed to it.

When the total rate of a non-GBR bearer (the sum of $Volume_{MeNB}$ and $Volume_{SgNB}$) is less than the value of the **gNBDUMacParamGroup.*DlGuaranteedRate*** parameter, the gNodeB preferentially allocates resources to the bearer to guarantee the minimum rate. **Figure 4-4** shows downlink data split with Option 3x.

4.1.2.1.3 Minimum Rate Guarantee for GBR Services

Minimum rate guarantee for GBR services is specific to one bearer.

- In NSA networking, the minimum rate to be guaranteed for GBR services is known as GBR and is indicated by the *GBR QoS Information* IE in an SGNB ADDITION REQUEST or SGNB MODIFICATION REQUEST message sent from the eNodeB to the gNodeB over the X2 interface.
- In SA networking, the minimum rate to be guaranteed for GBR services is known as guaranteed flow bit rate (GFBR) and is indicated by the *GBR QoS Flow Information* IE in a PDU SESSION RESOURCE SETUP REQUEST or PDU SESSION RESOURCE MODIFY REQUEST message sent from the 5GC to the gNodeB over the NG interface.

If there is data to be transmitted in the downlink buffer, a check is made in each TTI on whether the transmission traffic volume of a GBR bearer of a UE within a period of *T* is less than the product of the minimum guaranteed rate of GBR services and *T*:

- If so, radio resources are preferentially allocated to the bearer. This ensures that its rate reaches the minimum guaranteed rate of GBR services.

- If not, the priority calculated by basic scheduling is retained.

In NSA networking, *T* is 1000 ms. In SA networking, the value of *T* is that of **gNBQciBearer**.***AveragingWindowDur*** corresponding to the QCI level of a bearer.

4.1.2.1.4 Maximum Rate Limitation for GBR Services

Maximum rate limitation for GBR services is specific to one bearer. The gNodeB obtains the maximum rate allowed for GBR services from the eNodeB or the 5GC.

- In NSA networking, the maximum rate allowed for GBR services is known as MBR and is indicated by the *GBR QoS Information* IE in an SGNB ADDITION REQUEST or SGNB MODIFICATION REQUEST message sent from the eNodeB to the gNodeB over the X2 interface.
- In SA networking, the maximum rate allowed for GBR services is known as maximum flow bit rate (MFBR) and is indicated by the *GBR QoS Flow Information* IE in a PDU SESSION RESOURCE SETUP REQUEST or PDU SESSION RESOURCE MODIFY REQUEST sent from the 5GC to the gNodeB over the NG interface.

If there is data to be transmitted in the downlink buffer, a check is made in each TTI on whether the transmission traffic volume of a GBR bearer of a UE within a period of *T* is greater than the product of the maximum allowed rate of GBR services and *T*:

- If so, the data scheduling of the bearer must be suspended.
- If not, the priority calculated by basic scheduling is retained.

In NSA networking, *T* is 1000 ms. In SA networking, the value of *T* is that of **gNBQciBearer**.***AveragingWindowDur*** corresponding to the QCI level of a bearer.

4.1.2.1.5 ARP- and QCI-based Scheduling Priority Control

The gNodeB determines the downlink scheduling priority weighting factor of a bearer based on both the ARP and QCI, using limited system resources to provide differentiated services for different bearers.

An allocation and retention priority (ARP) is the priority of a bearer over other bearers for resource allocation and retention.

- In NSA networking, the gNodeB obtains the ARP from the *Allocation and Retention Priority* IE in the SGNB ADDITION REQUEST message sent by the eNodeB over the X2 interface.
- In SA networking, the gNodeB obtains the ARP from the *Allocation and Retention Priority* IE in the PDU SESSION RESOURCE SETUP REQUEST message sent by the core network over the NG interface.

After the ARP is obtained, the gNodeB can find the **gNBArp** MO where the **gNBArp**.***Arp*** parameter is set to this ARP value, and the **gNBArp**.***OperatorId*** and **gNBArp**.***Qci*** parameters can be configured for bearer mapping by operator, QCI, or ARP. The downlink scheduling priority weighting factor of a bearer is specified by the **gNBArp**.***DlSchPriWeightFactor*** parameter.

If some or all ARPs of a bearer corresponding to a QCI are not configured in the **gNBArp** MO, QCI-based scheduling priority control takes effect for the bearer. That is, the **gNBDUMacParamGroup**.***DlSchPriWeightFactor*** parameter value is used as the scheduling priority weighting factor.

Table 4-7 provides an example of ARP- and QCI-based downlink scheduling priorities. A bearer with a larger downlink scheduling priority weighting factor has a higher scheduling priority.

NOTE

In SA networking, when multiple QoS flows are mapped to one DRB:

- If the core network delivers the *Priority Level* IE for each QoS flow, the gNodeB selects the ARP corresponding to the 5QI with the highest priority (with the smallest *Priority Level* value) as the ARP of the bearer. If all 5QIs have the same priority, the ARP corresponding to the 5QI with the smallest QFI value is used as the ARP of the bearer.
- If the core network does not deliver the *Priority Level* IE for any QoS flow, the gNodeB selects the smallest ARP as the ARP of the bearer.

Table 4-7 Example of ARP- and QCI-based downlink scheduling priorities

QCI	ARP 1	ARP 2
6	200	2
7	1000	1000
8	800	720
9	700	600
Note: In the preceding table, the numbers except the QCIs in the first column indicate the downlink scheduling priority weighting factors.		

4.1.2.1.6 Scheduling Optimization for GBR and Non-GBR Services

On a network where downlink GBR and non-GBR services coexist, GBR services are preferentially guaranteed because they generally have higher priorities than non-GBR services. When air-interface resources in the cell are insufficient and there is a relatively large proportion of GBR services, there is a possibility that non-GBR services are not scheduled. To prevent this situation, scheduling optimization for GBR and non-GBR services can be enabled.

When the **GBR_NONGBR_SCH_OPT_SW** option of the **NRDUCellAlgoSwitch**.***ServiceDiffSwitch*** parameter is selected, scheduling optimization for GBR and non-GBR services takes effect. If the minimum rates of both downlink non-GBR and GBR services cannot be guaranteed, the priority of downlink non-GBR services is increased to the same level as that of downlink GBR services and the two types of services are guaranteed differentially by using the downlink scheduling priority weighting factor specified by **gNBDUMacParamGroup**.***DlSchPriWeightFactor***.

4.1.2.1.7 Downlink PDB-based Differentiated Scheduling

The packet delay budget (PDB) defines the maximum delay budget of data packets between a UE and the UPF. PDBs vary among services based on service characteristics. On the RAN side, PDB parameters can be delivered by the core network or configured by the base station using the **gNBQciBearer**.***PacketDelayBudget*** parameter. For details about the PDB, see **4.1.1.1.2 Configuring QCIs and Corresponding QoS Attributes**.

The core network packet delay budget (CNPDB) defines the maximum delay budget of data packets between the RAN and UPF. The CNPDB is configured by the base station using the **gNBQciBearer**.***CoreNetDelayBudget*** parameter. For details about CNPDB, see the descriptions related to **gNBQciBearer**.***CoreNetDelayBudget*** in *Layered QoS*.

The RAN-to-UE delay budget is determined by the result of PDB minus CNPDB.

Downlink PDB-based differentiated scheduling allows a gNodeB to compare the maximum and actual delays from the RAN to the UE. If the delay budget is about to be exceeded, the bearer scheduling priorities are adjusted to guarantee larger scheduling opportunities. This improves the PDB satisfaction rate and user experience.

Downlink PDB-based differentiated scheduling is enabled by setting the **gNBQciBearer**.***PdbDiffSchPolicy*** parameter to a value other than **OFF**. Differentiated scheduling policies vary if the **gNBQciBearer**.***PdbDiffSchPolicy*** parameter is set to different values. The recommended value of **gNBQciBearer**.***PdbDiffSchPolicy*** is related to the service type (specified by **gNBQciBearer**.***ServiceType***). For details, see **Table 4-8**.

Table 4-8 Definition and setting notes of different values for the downlink PDB-based differentiated scheduling policy

Value of gNBQciBearer.*PdbDiffSchPolicy*	Definition	Recommended Scenario
OFF	PDB-based differentiated scheduling is not applied to the bearer.	PDB-based differentiated scheduling is not required.
COMM_LOW_LATENCY_SERV	The differentiated scheduling policy applicable to common low-latency services is used for the bearer.	After this policy is used, the downlink rate of the bearer and the delay from the RAN to the UE may increase. It is recommended that this policy be configured for service types with related requirements.
NORMAL_SERV	The differentiated scheduling policy applicable to normal services is used for the bearer.	The **gNBQciBearer**.***ServiceType*** parameter is set to **NORMAL**.
SHORT_VIDEO_SERV	The differentiated scheduling policy applicable to short video services is used for the bearer.	The **gNBQciBearer**.***ServiceType*** parameter is set to **EMBB_SHORT_VIDEO**.

Value of gNBQciBearer.*PdbDiffSchPolicy*	Definition	Recommended Scenario
HIGH_RELIAB_SERV	The differentiated scheduling policy applicable to high-reliability services is used for the bearer.	The **gNBQciBearer**.***ServiceType*** parameter is set to **URLLC_HIGH_RELIABILITY_SERVICE** or **V2X_SERVICE**.
NORMAL_MOBILE_GAME_SERV	The differentiated scheduling policy applicable to normal mobile gaming services is used for the bearer.	The **gNBQciBearer**.***ServiceType*** parameter is set to **NORMAL** in mobile gaming service scenarios.

NOTE

The downlink PDB-based differentiated scheduling policy and the service type are not bound. Any downlink PDB-based scheduling policy can be configured for a service type.

When the **gNBQciBearer**.***PdbDiffSchPolicy*** parameter is set to **HIGH_RELIAB_SERV**, the maximum scheduling time budget can be configured by using the **gNBQciBearer**.***MaxSchTimeBudget*** parameter. A larger value of this parameter indicates a larger maximum scheduling time budget and a slower scheduling priority adjustment. A smaller value of this parameter results in the opposite effects.

Downlink PDB-based differentiated scheduling is applicable only in low frequency bands in SA networking.

4.1.2.2 QoS Guarantee for Uplink Scheduling

QoS guarantee for uplink scheduling differs from downlink scheduling as it must be performed by both UEs and the gNodeB. This is because the gNodeB cannot obtain the exact amount of data to be transmitted by UEs on each uplink bearer.

QoS Guarantee on the UE Side

The gNodeB calculates the scheduling priority of a UE based on the channel quality, historical transmission rate, and service QCI. For UEs that have obtained the uplink scheduling opportunity, a second scheduling is used to differentiate between bearers, and is performed based on the following information:

- Logical channel guaranteed rate
  - The logical channel guaranteed rate of a GBR bearer depends on the GBR configured on the core network, equaling its rounded down form to the nearest enumeration value or 8 kilobytes/s, whichever is higher.
  - The logical channel guaranteed rate is fixed at 8 kilobytes per second for non-GBR bearers.

  Logical channel guaranteed rate fulfillment is a key factor that affects the scheduling priority calculation of each logical channel.

- Logical channel priority

  The logical channel priority is used by UEs to manage the scheduling priority of each logical channel. It can be configured using the **gNBDUMacParamGroup**.***LogicalChnPriority*** parameter for each bearer.

- Logical channel-specific **Packet Delay Budget**

  Packet Delay Budget indicates the delay in scheduling data packets on the relevant bearer. It is a critical factor that affects the scheduling priority calculation of each logical channel.

  Similar to that in downlink scheduling, **Packet Delay Budget** can be configured using the **gNBQciBearer**.***PacketDelayBudget*** parameter.

  – In NSA networking, the default values of **Packet Delay Budget** for standardized QCIs are the same as those in the **Packet Delay Budget** column of **Table 4-1** and **Table 4-2**.

  – In SA networking, the default values of **Packet Delay Budget** for standardized QCIs are the same as those in the **Packet Delay Budget** column of **Table 4-3**, **Table 4-4**, and **Table 4-5**.

    If the AMF delivers this parameter during QoS flow establishment, the AMF-delivered parameter value is used. Otherwise, the **gNBQciBearer**.***PacketDelayBudget*** parameter value is used.

- Logical channel grouping

  5G protocols stipulate that the gNodeB supports eight logical channel groups. The group information can be configured using the **gNBDUMacParamGroup**.***LogicalChnGrpId*** parameter for each bearer.

The logical channel guaranteed rate, logical channel priority, logical channel-specific **Packet Delay Budget**, and logical channel grouping information are all delivered to UEs by using RRCReconfiguration messages.

QoS Guarantee on the gNodeB Side

The QoS guarantee on the gNodeB side determines the UE scheduling priority based on the channel quality, historical transmission rate, and service QCI.

The gNodeB calculates the uplink basic scheduling priority for each UE by using the following formula:

$$\mathrm{Priority} = \frac{\mathrm{eff}}{r} \times \gamma$$

In the formula:

- eff and r: For details, see *Uplink Scheduling*.
- γ: Indicates the uplink scheduling priority weighting factor of a specific service type. A larger factor indicates a higher uplink scheduling priority.
  – If the gNodeB does not obtain the RFSP index of a UE or the RFSP-relevant uplink scheduling priority coefficient (**gNBDURfspConfig**.***UlSchPriCoefficient***) is not configured for the UE: $\gamma = \gamma(\mathrm{QCI})$.
  – If the gNodeB obtains the RFSP index of a UE and the RFSP-relevant uplink scheduling priority coefficient (**gNBDURfspConfig**.***UlSchPriCoefficient***) is configured for the UE:

$\gamma = \gamma(QCI) \times Coeff_{RFSP}$. In this case, RFSP-based uplink scheduling priority control applies.

In the formula: $\gamma(QCI)$ indicates QCI-specific scheduling priority weighting (specified by **gNBDUMacParamGroup.*UlSchPriWeightFactor***) or QCI- and ARP-based scheduling priority weighting (specified by **gNBArp**.***UlSchPriWeightFactor***). For details about scheduling priority weighting for bearers with different QCIs or ARPs, see **4.1.2.2.5 ARP- and QCI-based Scheduling Priority Control**. $Coeff_{RFSP}$ indicates the uplink scheduling priority coefficient (specified by the **gNBDURfspConfig**.***UlSchPriCoefficient*** parameter) for UEs with the specified RFSP index. An operator can configure $Coeff_{RFSP}$ for specific UEs when desiring to customize uplink scheduling priorities for the UEs. For details about RFSP, see *Flexible User Steering*.

If a UE has multiple bearers, γ in the formula is configured using the weighting factor of the highest-priority logical channel in the highest-priority logical channel group.

For example, a UE has four bearers with QCI 9, QCI 1, QCI 5, and QCI 69, respectively. The configurations of **gNBDUMacParamGroup**.***LogicalChnGrpId*** referenced by the four bearers indicate that the bearers belong to three logical channel groups and the logical channel group for QCI 5 and QCI 69 has the highest priority. According to the **gNBDUMacParamGroup.*LogicalChnPriority*** configurations for QCI 5 and QCI 69, QCI 5 has a higher logical channel priority than QCI 69. Therefore, γ configured for the bearer with QCI 5 is used in the UE priority calculation formula. For detailed configurations, see **4.4.1.2 Using MML Commands**.

The scheduling type of a bearer is configured using the **gNBDUMacParamGroup**.***SchedulingType*** parameter.

NOTE

Services with the **gNBDUMacParamGroup**.***SchedulingType*** parameter set to **VOICE_SCH** are voice services, where differentiated scheduling priority processing is required for voice bearers. However, voice services are scheduled as common services in the following scenarios:

- A voice service is carried on a GBR bearer, and the uplink rate of the bearer is higher than twice the GBR.
- A voice service is carried on a non-GBR bearer, and the uplink rate of the bearer is higher than twice the **gNBDUMacParamGroup**.***UlGuaranteedRate*** parameter value.

QoS guarantee for uplink scheduling includes **4.1.2.2.1 UE-AMBR-based Rate Limitation for Non-GBR Services**, **4.1.2.2.2 Minimum Rate Guarantee for Non-GBR Services**, **4.1.2.2.3 Minimum Rate Guarantee for GBR Services**, **4.1.2.2.4 Maximum Rate Limitation for GBR Services**, **4.1.2.2.5 ARP- and QCI-based Scheduling Priority Control**, and **4.1.2.2.6 Scheduling Optimization for GBR and Non-GBR Services**.

4.1.2.2.1 UE-AMBR-based Rate Limitation for Non-GBR Services

Uplink UE-AMBR-based rate limitation for non-GBR services targets all uplink non-GBR bearers of a UE. Uplink UE-AMBR-based rate limitation can be enabled by selecting the **UL_AMBR_RATE_CTRL_SW** option of the **NRDUCellAlgoSwitch**.***ServiceDiffSwitch*** parameter. After this function takes effect, the gNodeB limits the total rate of all uplink non-GBR bearers of a UE so that the total rate does not exceed the uplink UE-AMBR.

- In NSA networking, the uplink UE-AMBR value is indicated by the *SgNB UE Aggregate Maximum Bit Rate* IE in an SGNB ADDITION REQUEST message sent from the eNodeB to the gNodeB over the X2 interface.
- In SA networking, the uplink UE-AMBR value is min{Value of the *UE Aggregate Maximum Bit Rate Uplink* IE, Sum of the values of the *PDU Session Aggregate Maximum Bit Rate* IE of all the PDU sessions established by a UE}. The sources of *UE Aggregate Maximum Bit Rate Uplink* and *PDU Session Aggregate Maximum Bit Rate* IEs are as follows:
  - *UE Aggregate Maximum Bit Rate Uplink* IE: carried in an INITIAL CONTEXT SETUP REQUEST or PDU SESSION RESOURCE SETUP REQUEST message sent from the 5GC to the gNodeB over the NG interface
  - *PDU Session Aggregate Maximum Bit Rate* IE: carried in *PDU Session Resource Setup Request Item* in a PDU SESSION RESOURCE SETUP REQUEST message or INITIAL CONTEXT SETUP REQUEST message sent from the 5GC to the gNodeB over the NG interface

Uplink UE-AMBR-based rate limitation for non-GBR services depends on the uplink BSR index.

- When the uplink BSR index of a UE is 0, the UE does not report the BSR index to the base station, and the base station does not schedule or process the data of the UE.
- If the uplink BSR index of a UE is greater than 0, a check is made in each TTI on whether the transmission traffic volume of all non-GBR bearers of the UE within a period of *T* is greater than the product of uplink UE-AMBR and *T*:
  - If so, data split and scheduling of each bearer are suspended.
  - If not, the priority calculated by basic scheduling is retained.

  In NSA networking, *T* is 500 ms. In SA networking, *T* is 3000 ms.

  In NSA DC scenarios, only the base station serving as the data split anchor can obtain the sum of the data volume carried on a bearer in both the MCG ($Volume_{MCG}$) and the SCG ($Volume_{SCG}$). The following describes the processing modes with Option 3 and Option 3x in NSA networking.

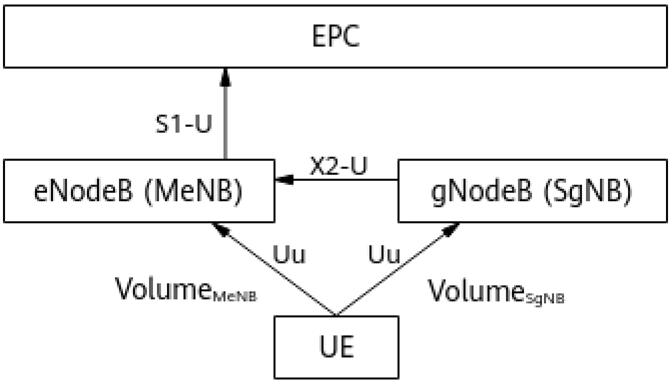
NOTE

There is no scenario where the uplink BSR index is less than 0.

With Option 3

Based on the value of the *UE Aggregate Maximum Bit Rate Uplink* IE, the gNodeB limits the rate for non-GBR bearers. When the $Volume_{SgNB}$ sum of all non-GBR bearers of a UE exceeds the UE-AMBR, the gNodeB stops uplink scheduling for the UE. **Figure 4-5** shows uplink data split with Option 3.

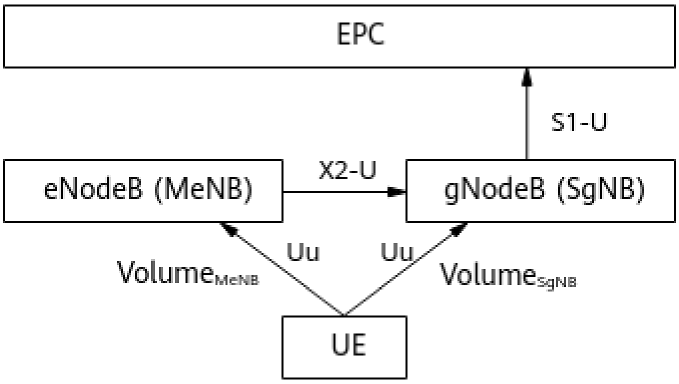
Figure 4-5 Uplink data split with Option 3



With Option 3x

The gNodeB limits the rate for non-GBR bearers based on the value of the *UE Aggregate Maximum Bit Rate Uplink* IE. When the sum of $Volume_{MeNB}$ and $Volume_{SgNB}$ of all non-GBR bearers of a UE exceeds the UE-AMBR, the gNodeB stops uplink scheduling for the UE. **Figure 4-6** shows uplink data split with Option 3x.

Figure 4-6 Uplink data split with Option 3x



4.1.2.2.2 Minimum Rate Guarantee for Non-GBR Services

Minimum rate guarantee for non-GBR services is used to ensure the minimum uplink rate for each non-GBR bearer.

- When the uplink BSR index of a UE is 0, the UE does not report the BSR index to the base station, and the base station does not schedule or process the data of the UE.
- When the uplink BSR index of a UE is greater than 0, a check is made in each TTI on whether the transmission traffic volume of a non-GBR bearer within a period of *T* is less than the product of the **gNBDUMacParamGroup.*UlGuaranteedRate*** parameter value and *T*.
  - If so, the scheduling priority of the bearer is increased to preferentially allocate radio resources to the bearer. This ensures that its rate reaches the value of the **gNBDUMacParamGroup.*UlGuaranteedRate*** parameter.
  - If not, the priority calculated by basic scheduling is retained.

In NSA networking, *T* is 500 ms. In SA networking, *T* is 3000 ms.

NOTE

- There is no scenario where the uplink BSR index is less than 0.
- If the **gNBDUMacParamGroup.*UlGuaranteedRate*** parameter is set to **0**, rate guarantee for uplink non-GBR services is disabled. Generally, non-GBR services have a lower priority and a larger traffic volume than GBR services. Therefore, enabling rate guarantee for uplink non-GBR services may consume more resources. It is recommended that this function be used only when the minimum rate of uplink non-GBR services needs to be guaranteed.

In NSA DC scenarios, only the base station serving as the data split anchor can obtain the sum of the data volume carried on a bearer in both the MCG ($Volume_{MCG}$) and SCG ($Volume_{SCG}$). The following describes the processing modes with Option 3 and Option 3x in NSA networking.

With Option 3

The gNodeB guarantees the minimum rate for traffic distributed to it. If the current rate is lower than the value of the **gNBDUMacParamGroup.*UlGuaranteedRate*** parameter for a bearer, the gNodeB preferentially allocates resources for the bearer to ensure that the minimum rate is reached. **Figure 4-5** shows uplink data split with Option 3.

With Option 3x

The gNodeB calculates the sum of $Volume_{MeNB}$ and $Volume_{SgNB}$ of a bearer. If the calculated rate is lower than the value of the **gNBDUMacParamGroup.*UlGuaranteedRate*** parameter for the bearer, the gNodeB preferentially allocates resources for the bearer to ensure that the minimum rate is reached in best effort mode. **Figure 4-6** shows uplink data split with Option 3x.

4.1.2.2.3 Minimum Rate Guarantee for GBR Services

Minimum rate guarantee for GBR services is specific to one bearer.

- In NSA networking, the minimum rate to be guaranteed for GBR services is known as GBR and is indicated by the *GBR QoS Information* IE in an SGNB ADDITION REQUEST or SGNB MODIFICATION REQUEST message sent from the eNodeB to the gNodeB over the X2 interface.
- In SA networking, the minimum rate to be guaranteed for GBR services is known as GFBR and is indicated by the *GBR QoS Flow Information* IE in a PDU SESSION RESOURCE SETUP REQUEST or PDU SESSION RESOURCE MODIFY REQUEST sent from the 5GC to the gNodeB over the NG interface.

Minimum rate guarantee for uplink GBR services depends on the uplink BSR index.

- When the uplink BSR index of a UE is 0, the UE does not report the BSR index to the base station, and the base station does not schedule or process the data of the UE.
- When the uplink BSR index of a UE is greater than 0, a check is made in each TTI on whether the transmission traffic volume of a GBR bearer of the UE within a period of *T* is less than the product of the minimum guaranteed rate for GBR services and *T*.

  - If so, radio resources are preferentially allocated to the bearer to ensure that its rate reaches the configured value.
  - If not, the priority calculated by basic scheduling is retained.

  In NSA networking, *T* is fixed at 500 ms. In SA networking, the value of *T* is that of **gNBQciBearer**.***AveragingWindowDur*** corresponding to the QCI level of a bearer.

NOTE

There is no scenario where the uplink BSR index is less than 0.

4.1.2.2.4 Maximum Rate Limitation for GBR Services

Maximum rate limitation for GBR services is specific to one bearer.

- In NSA networking, the maximum rate allowed for GBR services is known as MBR and is indicated by the *GBR QoS Information* IE in an SGNB ADDITION REQUEST or SGNB MODIFICATION REQUEST message sent from the eNodeB to the gNodeB over the X2 interface.
- In SA networking, the maximum rate allowed for GBR services is known as MFBR and is indicated by the *GBR QoS Flow Information* IE in a PDU SESSION RESOURCE SETUP REQUEST or PDU SESSION RESOURCE MODIFY REQUEST sent from the 5GC to the gNodeB over the NG interface.

Maximum rate limitation for uplink GBR services depends on the uplink BSR index.

- When the uplink BSR index of a UE is 0, the UE does not report the BSR index to the base station, and the base station does not schedule or process the data of the UE.
- When the uplink BSR index of a UE is greater than 0, a check is made in each TTI on whether the transmission traffic volume of a GBR bearer of the UE within a period of *T* is greater than the product of the maximum allowed rate for GBR services and *T*.
  - If so, the data scheduling of the bearer must be suspended.
  - If not, the priority calculated by basic scheduling is retained.

  In NSA networking, *T* is fixed at 500 ms. In SA networking, the value of *T* is that of **gNBQciBearer**.***AveragingWindowDur*** corresponding to the QCI level of a bearer.

NOTE

There is no scenario where the uplink BSR index is less than 0.

4.1.2.2.5 ARP- and QCI-based Scheduling Priority Control

The gNodeB determines the uplink scheduling priority weighting factor of a bearer based on both the ARP and QCI, using limited system resources to provide differentiated services for different bearers.

An ARP is the priority of a bearer over other bearers for resource allocation and retention.

- In NSA networking, the gNodeB obtains the ARP from the *Allocation and Retention Priority* IE in the SGNB ADDITION REQUEST message sent by the eNodeB over the X2 interface.

- In SA networking, the gNodeB obtains the ARP from the *Allocation and Retention Priority* IE in the PDU SESSION RESOURCE SETUP REQUEST message sent by the core network over the NG interface.

After the ARP is obtained, the gNodeB can find the **gNBArp** MO where the **gNBArp**.***Arp*** parameter is set to this ARP value, and the **gNBArp**.***OperatorId*** and **gNBArp**.***Qci*** parameters can be configured for bearer mapping by operator, QCI, or ARP. The uplink scheduling priority weighting factor of a bearer is specified by the **gNBArp**.***UlSchPriWeightFactor*** parameter.

If some or all ARPs of a bearer corresponding to a QCI are not configured in the **gNBArp** MO, QCI-based scheduling priority control takes effect for the bearer. That is, the **gNBDUMacParamGroup**.***UlSchPriWeightFactor*** parameter value is used as the scheduling priority weighting factor.

Table 4-9 provides an example of ARP- and QCI-based uplink scheduling priorities. A bearer with a larger uplink scheduling priority weighting factor has a higher scheduling priority.

NOTE

In SA networking, when multiple QoS flows are mapped to one DRB:

- If the core network delivers the *Priority Level* IE for each QoS flow, the gNodeB selects the ARP corresponding to the 5QI with the highest priority (with the smallest *Priority Level* value) as the ARP of the bearer. If all 5QIs have the same priority, the ARP corresponding to the 5QI with the smallest QFI value is used as the ARP of the bearer.
- If the core network does not deliver the *Priority Level* IE for any QoS flow, the gNodeB selects the smallest ARP as the ARP of the bearer.

Table 4-9 Example of ARP- and QCI-based uplink scheduling priorities

QCI	ARP 1	ARP 2
6	200	2
7	1000	1000
8	800	720
9	700	600
Note: In the preceding table, the numbers except the QCIs in the first column indicate the uplink scheduling priority weighting factors.		

4.1.2.2.6 Scheduling Optimization for GBR and Non-GBR Services

On a network where uplink GBR and non-GBR services coexist, GBR services are preferentially guaranteed because they generally have higher priorities than non-GBR services. When air-interface resources in the cell are insufficient and there is a relatively large proportion of GBR services, there is a possibility that non-GBR services are not scheduled. To prevent this situation, scheduling optimization for GBR and non-GBR services can be enabled.

When the **GBR_NONGBR_SCH_OPT_SW** option of the **NRDUCellAlgoSwitch**.***ServiceDiffSwitch*** parameter is selected, scheduling optimization for GBR and non-GBR services takes effect. If the minimum rates of both uplink non-GBR and GBR services cannot be guaranteed, the priority of uplink non-GBR services is increased to the same level as that of uplink GBR services and the two types of services are guaranteed differentially by using the uplink scheduling priority weighting factor specified by **gNBDUMacParamGroup**.***UlSchPriWeightFactor***.

4.1.2.3 QoS Guarantee for Non-Scheduling Procedure

In non-scheduling procedures, notification control can be used to implement QoS guarantee.

When there are uplink or downlink GBR services on a network, it is challenging to achieve the GFBR and delay goals for the GBR services in certain scenarios, such as those with weak coverage or strong interference. Excessive RB resources will be consumed if the GFBR and delay goals are retained, thereby affecting other UEs. In this case, notification control is introduced.

This function is controlled by the **NOTIF_CTRL_SW** option of the **gNB5qiConfig**.***Nr5qiAlgoSwitch*** parameter. After the notification control function is enabled, the gNodeB periodically checks the guarantee status of the uplink GFBR, downlink GFBR, and downlink packet delay budget (PDB) of the GBR QoS flow and reports the guarantee status to the core network. The check interval can be configured through the **gNodeBParam**.***NotifCtrlCheckInterval*** parameter. During the guarantee status check, the GFBR guarantee status is always checked, while the PDB guarantee status is checked only when the **gNBQciBearer**.***QncArrangeType*** parameter is set to **DELAY_SENSITIVE**.

The gNodeB reports a message either indicating an estimated guarantee success or estimated guarantee failure. The core network can analyze service quality and adjust the GFBR based on the message reported by the gNodeB. For example, if the message indicates an estimated guarantee failure, the core network can decrease the levels of GFBRs and delay according to policies. The processing on the core network depends on the functions implemented on the core network. If the guarantee status is checked for both the GFBRs and PDB, the gNodeB reports a message indicating an estimated guarantee failure if either of them cannot be guaranteed.

The resource range for estimation by the gNodeB can be configured through the **gNBQciBearer**.***NotifCtrlResFactor*** parameter. The parameter value is in percentage. If this parameter is set to 100, the estimation is performed based on all remaining resources that can be allocated to the bearer in the cell. If this parameter is set to a value below 100 (such as 80), the estimation is performed based on 80% of the remaining resources that can be allocated to the bearer in the cell.

The notification control function works only in low frequency bands in SA networking.

4.2 Network Analysis

4.2.1 Benefits

QoS guarantee for uplink scheduling can meet the diverse requirements of UEs and maximize uplink capacity while helping ensure uplink coverage and resolve uplink interference.

QoS guarantee for downlink scheduling can meet the diverse requirements of UEs and maximize downlink capacity while ensuring downlink coverage. In addition, downlink PDB-based differentiated scheduling can meet the diverse requirements of UEs and improve the downlink PDB satisfaction rate.

In QoS guarantee for non-scheduling procedure, notification control enables the core network to meet the diverse requirements of GBR service UEs.

4.2.2 Impacts

The impacts between this function and other functions are described in **4.3.2.3 Function Impacts**.

A scheduling priority weighting factor will affect the scheduling priority of the service with a specific QCI. A larger factor results in a higher scheduling priority and better user experience. A smaller factor leads to the opposite effect. Follow the suggestions listed below on configuring scheduling priority weighting factors for different QCIs in a cell. Otherwise, services with low scheduling priority weighting factors may not be scheduled, affecting user experience.

- It is recommended that the maximum scheduling priority weighting factor be no more than 200 times the minimum one.
- If the maximum scheduling priority weighting factor is over 200 times the minimum one due to special reasons, it is recommended that minimum rate guarantee for non-GBR services be enabled. In this way, the basic experience of services with low scheduling priority weighting factors can be ensured.

The downlink and uplink scheduling priority weighting factors are Y_{conn} in **4.1.2.1 QoS Guarantee for Downlink Scheduling** and Y in **4.1.2.2 QoS Guarantee for Uplink Scheduling**, respectively. For details about minimum rate guarantee for non-GBR services in the downlink and uplink, see **4.1.2.1.2 Minimum Rate Guarantee for Non-GBR Services** and **4.1.2.2.2 Minimum Rate Guarantee for Non-GBR Services**, respectively.

In user rate control, if the configured AMBR is less than the rate supported by the air interface, the cell throughput cannot reach the actual amount supported by the cell, decreasing the cell capacity. If UEs are located at the cell edge but a large GBR or GFBR is configured, resources are preferentially allocated to this type of UE to ensure their minimum rate, which decreases cell throughput. Due to different UE distributions in a cell, the cell throughput varies as the scheduling priority changes.

In RFSP-based scheduling priority control:

- In single-UE scenarios, the scheduling priority coefficient does not affect the UE throughput.
- In multi-UE scenarios, a smaller scheduling priority coefficient indicates a lower UE throughput, and a larger scheduling priority coefficient indicates a higher UE throughput.

With ARP- and QCI-based scheduling priority control:

- In single-UE scenarios, the scheduling priority weighting factor does not affect the UE throughput.
- In multi-UE scenarios, a smaller scheduling priority weighting factor leads to a lower UE throughput whereas a higher scheduling priority weighting factor leads to a higher UE throughput.

When the **PDU_SESSION_MOD_COMPAT_OPT_SW** option of the **gNodeBParam**.***CompatibilityAlgoSwitch*** parameter is selected and switching between dynamic 5QIs and non-dynamic 5QIs occurs, the value of **N.PDUSession.AMF.Mod.Succ** decreases.

When the **GBR_ADMIT_BY_23501_SW** option of the **gNodeBParam**.***CompatibilityAlgoSwitch*** parameter is selected and the 5QI-specific service type of a QoS flow is inconsistent with the core-network-delivered one, the value of **N.QosFlow.Est.Succ** decreases.

If the **GBR_NONGBR_SCH_OPT_SW** option of the **NRDUCellAlgoSwitch**.***ServiceDiffSwitch*** parameter is selected and both non-GBR and GBR services are running on the network, non-GBR services are allocated more resources and GBR services are allocated fewer resources during scheduling. This increases the traffic volume of non-GBR services, but decreases that of GBR services in congestion scenarios.

After downlink PDB-based differentiated scheduling is enabled, the delay satisfaction rate increases for services on which PDB-based differentiated scheduling has taken effect but may decrease for other services in the relevant cells.

After downlink PDB-based differentiated scheduling is enabled:

- In single-UE scenarios, the scheduling priority coefficient does not affect the UE throughput.
- In multi-UE scenarios, a smaller scheduling priority coefficient indicates a lower UE throughput, and a larger scheduling priority coefficient indicates a higher UE throughput.

After notification control is enabled, the core network may decrease the levels of GFBRs and delay according to policies if the gNodeB reports a message indicating an estimated guarantee failure. The processing on the core network depends on the functions implemented on the core network.

4.3 Requirements

4.3.1 Licenses

There are no license requirements for basic functions.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Notification Control	**NOTIF_CTRL_SW** option of the **gNB5qiConfig**.***Nr5qiAlgoSwitch*** parameter	*QoS Management*	The **gNodeBParam**.***NotifCtrlCheckInterval*** parameter takes effect only when the notification control function is enabled.
Low-frequency TDD	Notification Control	**NOTIF_CTRL_SW** option of the **gNB5qiConfig**.***Nr5qiAlgoSwitch*** parameter	*QoS Management*	The **gNBQciBearer**.***NotifCtrlResFactor*** parameter takes effect only when the notification control function is enabled.
Low-frequency TDD	Downlink PDB-based differentiated scheduling	**gNBQciBearer**.***PdbDiffSchPolicy***	*QoS Management*	The **gNBQciBearer**.***MaxSchTimeBudget*** parameter takes effect only when the **gNBQciBearer**.***PdbDiffSchPolicy*** parameter is set to **HIGH_RELIAB_SERV**.
Low-frequency TDD High-frequency TDD	PDU Session Resource Modify compatibility optimization	**PDU_SESSION_MOD_COMPAT_OPT_SW** option of the **gNodeBParam**.***CompatibilityAlgoSwitch*** parameter	*QoS Management*	PDU Session Resource Modify QCI mapping optimization can be enabled only when PDU Session Resource Modify compatibility optimization is enabled.

4.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NR 5G QoS Indicator	**gNB5qiConfig**.***Nr5qi***	*QoS Management*	The notification control function must be disabled when the **gNB5qiConfig**.***Nr5qi*** parameter is set to **1**, **2**, **5**, **65**, **66**, **67**, or **69**.
Low-frequency TDD	PDU Set identification	**gNBQciBearer**.***IdentificationMode*** set to **R18_PDU_SET_MODE**	*Layered QoS*	PDU Set identification cannot be enabled when the **gNBQciBearer**.***Qci*** parameter is set to **1**, **2**, or **5**.
High-frequency TDD	QCI-specific uplink data compression	**QCI_UDC_SW** option of the **gNBQciBearer**.***PdcpAlgoSwitch*** parameter	*Ultra Uplink (High-Frequency TDD)*	When the **gNBQciBearer**.***Qci*** parameter is set to **1**, **65**, or **66**, the **QCI_UDC_SW** option of the **gNBQciBearer**.***PdcpAlgoSwitch*** parameter cannot be selected.
Low-frequency TDD	QCI-specific uplink data compression	**QCI_UDC_SW** option of the **gNBQciBearer**.***PdcpAlgoSwitch*** parameter	*Turbo Uplink (Low-Frequency TDD)*	When the **gNBQciBearer**.***Qci*** parameter is set to **1**, **65**, or **66**, the **QCI_UDC_SW** option of the **gNBQciBearer**.***PdcpAlgoSwitch*** parameter cannot be selected.

4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	User PRB control for FWA	**PRB_CTRL_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter	*FWA*	When the uplink or downlink scheduling priority weighting factor is set to a rather small value, if the PRB usage exceeds the average RB proportion threshold after user PRB control for FWA is enabled, lowering the scheduling priority may cause loss of fairness among UEs.
Low-frequency TDD High-frequency TDD	Uplink experience-based scheduling	**UL_EXP_BASED_SCH_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter	*FWA*	When the uplink guaranteed rate and uplink experience-based scheduling take effect for different users at the same time, resources are preferentially allocated to the user with the uplink guaranteed rate. When the uplink guaranteed rate and uplink experience-based scheduling simultaneously take effect for a user, the user rate improvement is primarily attributed to the uplink guaranteed rate.
Low-frequency TDD	Scheduling based on delay-sensitive requirements	**gNBDUSchParamGroup**.***CycleTime***	*URLLC*	When both scheduling based on delay-sensitive requirements and downlink PDB-based differentiated scheduling are enabled, only downlink PDB-based differentiated scheduling takes effect.
Low-frequency TDD	RB-usage-based rate policy	**RB_USAGE_BASED_RATE_POL_SW** option of the **NRDUCellRes**.***SchedulingPolicySwitch*** parameter	*QCI-based Resource Control*	When the RB-usage-based rate policy (controlled by the **RB_USAGE_BASED_RATE_POL_SW** option of the **NRDUCellRes**.***SchedulingPolicy-Switch*** parameter) takes effect, the minimum rate of GBR services may fail to be guaranteed.
Low-frequency TDD	RB-usage-based rate policy	**RB_USAGE_BASED_RATE_POL_SW** option of the **NRDUCellRes**.***SchedulingPolicySwitch*** parameter	*QCI-based Resource Control*	When the RB-usage-based rate policy (controlled by the **RB_USAGE_BASED_RATE_POL_SW** option of the **NRDUCellRes**.***SchedulingPolicy-Switch*** parameter) takes effect, the minimum rate of non-GBR services may fail to be guaranteed.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Downlink inter-gNodeB CA	**INTER_GNODEB_CA_SW** option of the **NRDUCellAlgoSwitch.*CaAlgoSwitch*** parameter	*CA Experience Improvement*	Consider a downlink inter-gNodeB CA scenario where the version of the gNodeB serving the PCell is V100R022C10SPC050 or later and the version of the gNodeB serving the SCell is V100R022C10SPC010 or earlier. If the **gNBDUMacParamGroup.*DlSchPriWeightFactor*** or **gNBArp.*DlSchPriWeightFactor*** parameter (**Downlink Scheduling Priority Weight Factor**) is greater than **1000**, the value of $\gamma(QCI) \times Coeff_{RFSP}$ (product of the GUI values) may be greater than **65535000**. In this case, due to the version restriction of the gNodeB serving the SCell, the SCell's effective downlink scheduling priority weighting factor (which should use the PCell configuration in a downlink inter-gNodeB CA scenario) is actually less than the PCell's configured value. When it comes to concurrent scheduling of multiple services, the SCell delivers lower data rates to CA UEs than expected based on the PCell's configured scheduling priority weighting factor. For details about $\gamma(QCI) \times Coeff_{RFSP}$, see **4.1.2.1 QoS Guarantee for Downlink Scheduling**.
Low-frequency TDD	Flexible experience differentiation	**FLEX_EXP_DIFF_SW** option of the **NRDUCellFeatureSw.*QciBasedDiffServExpSw*** parameter	*Service-differentiated Experience Guarantee*	When a UE performs GBR services and "flexible experience differentiation" is enabled, the UE's uplink or downlink peak data rate takes the value of the target uplink or downlink data rate calculated in "flexible experience differentiation" if the UE's MBR or MFBR is greater than the aforementioned calculated target uplink or downlink data rate.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Flexible experience differentiation	**FLEX_EXP_DIFF_SW** option of the **NRDUCellFeatureSw**.***QciBasedDiffServExpSw*** parameter	*Service-differentiated Experience Guarantee*	When a UE performs GBR services and "flexible experience differentiation" is enabled, the UE's uplink or downlink peak data rate takes the value of the GBR or GFBR configured on the core network if the UE's GBR or GFBR is greater than the target uplink or downlink data rate calculated in "flexible experience differentiation". As a result, the UE's uplink or downlink data rate may not be consistent with the flexible experience differentiation configuration, which may lead to an actual data rate lower than the minimum guaranteed rate.
Low-frequency TDD	Flexible experience differentiation	**FLEX_EXP_DIFF_SW** option of the **NRDUCellFeatureSw**.***QciBasedDiffServExpSw*** parameter	*Service-differentiated Experience Guarantee*	After "flexible experience differentiation" is enabled, a UE's uplink peak data rate takes the value of the **gNBDUMacParamGroup**.***UlGuaranteedRate*** parameter if this parameter is set to a non-zero value which exceeds the target uplink data rate calculated in "flexible experience differentiation". As a result, the UE's uplink data rate may not be consistent with the flexible experience differentiation configuration, which may lead to an actual data rate lower than the minimum guaranteed rate.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Flexible experience differentiation	**FLEX_EXP_DIFF_SW** option of the **NRDUCellFeatureSw**.***QciBasedDiffServExpSw*** parameter	*Service-differentiated Experience Guarantee*	After "flexible experience differentiation" is enabled, a UE's downlink peak data rate takes the value of the **gNBDUMacParamGroup.*DlGuaranteedRate*** parameter if this parameter is set to a non-zero value which exceeds the target downlink data rate calculated in "flexible experience differentiation". As a result, the UE's downlink data rate may not be consistent with the flexible experience differentiation configuration, which may lead to an actual data rate lower than the minimum guaranteed rate.
Low-frequency TDD	Flexible experience differentiation	**FLEX_EXP_DIFF_SW** option of the **NRDUCellFeatureSw**.***QciBasedDiffServExpSw*** parameter	*Service-differentiated Experience Guarantee*	After both "uplink UE-AMBR-based rate limitation for non-GBR services" and "flexible experience differentiation" are enabled, the UE's uplink peak data rate takes the value of the target uplink data rate calculated in "flexible experience differentiation" if the uplink UE-AMBR used for "uplink UE-AMBR-based rate limitation for non-GBR services" is greater than the aforementioned calculated target uplink data rate.
Low-frequency TDD	Flexible experience differentiation	**FLEX_EXP_DIFF_SW** option of the **NRDUCellFeatureSw**.***QciBasedDiffServExpSw*** parameter	*Service-differentiated Experience Guarantee*	After both "downlink UE-AMBR-based rate limitation for non-GBR services" and "flexible experience differentiation" are enabled, the UE's downlink peak data rate takes the value of the target downlink data rate calculated in "flexible experience differentiation" if the downlink UE-AMBR used for "downlink UE-AMBR-based rate limitation for non-GBR services" is greater than the aforementioned calculated target downlink data rate.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

This function does not depend on RF modules.

4.3.4 Others

Basic notification control in non-scheduling procedures requires the core network's support. Specifically, QoS flow information delivered by the core network must contain the **Notification** field.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

The NR system supports parameter configurations for standardized QCIs defined in section 6.1.7 "Standardized QoS characteristics" of 3GPP TS 23.203 V15.4.0 and for standardized 5QIs defined in section 5.7.4 "Standardized 5QI to QoS characteristics mapping" of 3GPP TS 23.501 V15.4.0. The NR system also supports parameter configurations for extended QCIs and extended 5QIs, in the same way as those for standardized QCIs and standardized 5QIs, respectively.

- Standardized QCIs (QCI 1 to QCI 9, QCI 65 to QCI 67, QCI 69 to QCI 76, QCI 79 to QCI 80, and QCI 82 to QCI 86) can be created by the system by default without the need for parameter configurations. Therefore, activation is not involved. **Table 4-11** lists the related parameters used for optimization. This table does not describe parameters related to cell establishment.
- Extended QCIs cannot be created by the system by default and require configurations of related parameters. **Table 4-10** and **Table 4-11** list the

related parameters used for activation and optimization, respectively. This table does not describe parameters related to cell establishment. When an extended QCI is manually created, related transmission configurations are also required to set up the mapping between the QCI and the UE data type (**UDT.*UDTNO***). For details, see *Transmission Resource Management*.

Table 4-10 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
QoS Class Identifier	**gNBQciBearer**.***Qci***	The standardized QCIs that have been created by the system by default do not need to be configured. Configure extended QCIs based on the network plan.
QoS Class Identifier	**gNB5qiConfig**.***Qci***	
QoS Class Identifier	**NRCellQciBearer**.***Qci***	
NR DU Cell ID	**NRDUCellQciBearer**.***NrDuCellId***	
QoS Class Identifier	**NRDUCellQciBearer**.***Qci***	
MAC Parameter Group ID	**NRDUCellQciBearer**.***MacParamGroupId***	
UM RLC Parameter Group ID	**NRDUCellQciBearer**.***UmRlcParamGroupId***	
AM RLC Parameter Group ID	**NRDUCellQciBearer**.***AmRlcParamGroupId***	
AM PDCP Parameter Group ID	**NRCellQciBearer**.***AmPdcpParamGroupId***	
UM PDCP Parameter Group ID	**NRCellQciBearer**.***UmPdcpParamGroupId***	
NR DU Cell ID	**NRDUCellQciBearer**.***NrDuCellId***	Set this parameter based on the network plan.
NR Cell ID	**NRCellQciBearer**.***NrCellId***	
QoS Class Identifier	**NRCellOpQciBearer**.***QCI***	
Operator ID	**NRCellOpQciBearer**.***OperatorId***	

Parameter Name	Parameter ID	Setting Notes
AM PDCP Parameter Group ID	**NRCellOpQciBearer**.***AmPdcpParamGroupId***	
UM PDCP Parameter Group ID	**NRCellOpQciBearer**.***UmPdcpParamGroupId***	
RLC Mode	**NRCellOpQciBearer**.***RlcMode***	
PDB Differentiated Scheduling Policy	**gNBQciBearer**.***PdbDiffSchPolicy***	Set this parameter based on **4.1.2.1.7 Downlink PDB-based Differentiated Scheduling** and the network plan.
Maximum Scheduling Time Budget	**gNBQciBearer**.***MaxSchTimeBudget***	Retain the default value. A larger value of this parameter indicates a larger maximum scheduling time budget and a slower scheduling priority adjustment. A smaller value of this parameter results in the opposite effects.
NR 5QI Algorithm Switch	**gNB5qiConfig**.***Nr5qiAlgoSwitch***	Select the **NOTIF_CTRL_SW** option if the notification control function is required.
Notification Control Check Interval	**gNodeBParam**.***NotifCtrlCheckInterval***	Retain the default value.
Notification Control Resource Factor	**gNBQciBearer**.***NotifCtrlResFactor***	Set this parameter based on the network plan. A larger value of this parameter results in a larger number of active UEs and larger traffic volume with the specified 5QI. A smaller value of this parameter produces the opposite effects.

Parameter Name	Parameter ID	Setting Notes
QNC Arrange Type	**gNBQciBearer**.***QncArrangeType***	Set this parameter based on the network plan. Set this parameter to **DELAY_SENSITIVE** if the gNodeB is required to check the PDB guarantee status. In other scenarios, set this parameter to a value other than **DELAY_SENSITIVE**.
Compatibility Algorithm Switch	**PDU_SESSION_MOD_QCI_MAP_OPT_SW** option of the **gNodeBParam**.***CompatibilityAlgoSwitch*** parameter	It is recommended that this option be selected.

Table 4-11 Parameters used for optimization

Parameter Name	Parameter ID	Setting Notes
QoS Class Identifier	**gNBQciBearer**.***Qci***	The standardized QCIs that have been created by the system by default do not need to be configured. Configure extended QCIs based on the network plan.
Averaging Window Duration	**gNBQciBearer**.***AveragingWindowDur***	Retain the default value.
Packet Delay Budget	**gNBQciBearer**.***PacketDelayBudget***	Retain the default value.
Priority Level	**gNBQciBearer**.***PriorityLevel***	Retain the default value.
Counter Object Subscription Indicator	**gNBQciBearer**.***CounterObjSubscriptionInd***	When a QCI is added to the network, set this parameter to **YES** if subscription to performance counters of this QCI is required.
NR 5G QoS Indicator	**gNB5qiConfig**.***Nr5qi***	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
QoS Class Identifier	**gNB5qiConfig**.***Qci***	The standardized QCIs that have been created by the system by default do not need to be configured. Configure extended QCIs based on the network plan.
AS Reflective QoS Switch	**gNB5qiConfig**.***AsReflectiveQosSwitch***	Set this parameter to **ON(On)** when multiple QoS flows are mapped onto the same DRB.
Counter Object Subscription Indicator	**gNB5qiConfig**.***CounterObjSubscriptionInd***	When a 5QI is added to the network, set this parameter to **YES** if subscription to performance counters of this 5QI is required.
NR Cell ID	**NRCellQciBearer**.***NrCellId***	Set this parameter based on the network plan.
QoS Class Identifier	**NRCellQciBearer**.***Qci***	The standardized QCIs that have been created by the system by default do not need to be configured. Configure extended QCIs based on the network plan.
AM PDCP Parameter Group ID	**NRCellQciBearer**.***AmPdcpParamGroupId***	Set this parameter to its recommended value.
UM PDCP Parameter Group ID	**NRCellQciBearer**.***UmPdcpParamGroupId***	Set this parameter to its recommended value.

Parameter Name	Parameter ID	Setting Notes
RLC Mode	**NRCellQciBearer**.***RlcMode***	This parameter can be set by the system by default. The default value is as follows: ● For bearers with a QCI of 4, 5, 6, 8, 9, 69, 70, 71, 72, 73, 74, 76, or 80, a newly deployed gNodeB automatically uses **AM** as the value of **NRCellQciBearer**.***RlcMode***. ● For bearers with a QCI of 1, 2, 3, 7, 65, 66, 67, 75, 79, 82, 83, 84, 85, or 86, a newly deployed gNodeB automatically uses **UM** as the value of **NRCellQciBearer**.***RlcMode***.
NR DU Cell ID	**NRDUCellQciBearer**.***NrDuCellId***	Set this parameter based on the network plan.
QoS Class Identifier	**NRDUCellQciBearer**.***Qci***	The standardized QCIs that have been created by the system by default do not need to be configured. Configure extended QCIs based on the network plan.
MAC Parameter Group ID	**NRDUCellQciBearer**.***MacParamGroupId***	Set this parameter to its recommended value.
UM RLC Parameter Group ID	**NRDUCellQciBearer**.***UmRlcParamGroupId***	Set this parameter to its recommended value.
AM RLC Parameter Group ID	**NRDUCellQciBearer**.***AmRlcParamGroupId***	Set this parameter to its recommended value.
MAC Parameter Group ID	**gNBDUMacParamGroup**.***MacParamGroupId***	Set this parameter based on the network plan.
Downlink Guaranteed Rate	**gNBDUMacParamGroup**.***DlGuaranteedRate***	Set this parameter based on the downlink rate to be guaranteed for non-GBR bearers.
Uplink Guaranteed Rate	**gNBDUMacParamGroup**.***UlGuaranteedRate***	Set this parameter based on the uplink rate to be guaranteed for non-GBR bearers.

Parameter Name	Parameter ID	Setting Notes
Downlink Scheduling Priority Weight Factor	**gNBDUMacParamGroup**.***DlSchPriWeightFactor***	Retain the default value. Y_{conn} is subject to the settings of this parameter. For details about the impacts of and suggestions on the configuration of Y_{conn}, see **4.2.2 Impacts**.
Uplink Scheduling Priority Weight Factor	**gNBDUMacParamGroup**.***UlSchPriWeightFactor***	Retain the default value. Y is subject to the settings of this parameter. For details about the impacts of and suggestions on the configuration of Y, see **4.2.2 Impacts**.
Logical Channel Group ID	**gNBDUMacParamGroup**.***LogicalChnGrpId***	Set this parameter based on the network plan.
Logical Channel Priority	**gNBDUMacParamGroup**.***LogicalChnPriority***	Retain the default value.
Scheduling Type	**gNBDUMacParamGroup**.***SchedulingType***	Retain the default value.
Operator ID	**gNBArp**.***OperatorId***	Set this parameter based on the network plan.
QoS Class Identifier	**gNBArp**.***Qci***	Set this parameter based on the network plan.
Allocation and Retention Priority	**gNBArp**.***Arp***	Set this parameter based on the network plan.
Downlink Scheduling Priority Weight Factor	**gNBArp**.***DlSchPriWeightFactor***	Set this parameter based on the network plan. Y_{conn} is subject to the settings of this parameter. For details about the impacts of and suggestions on the configuration of Y_{conn}, see **4.2.2 Impacts**.

Parameter Name	Parameter ID	Setting Notes
Uplink Scheduling Priority Weight Factor	**gNBArp**.***UlSchPriWeightFactor***	Set this parameter based on the network plan. Y is subject to the settings of this parameter. For details about the impacts of and suggestions on the configuration of Y, see **4.2.2 Impacts**.
NR DU Cell ID	**NRDUCellAlgoSwitch**.***NrDuCellId***	Set this parameter based on the network plan.
Service Differentiation Switch	**NRDUCellAlgoSwitch**.***ServiceDiffSwitch***	Set the **DL_AMBR_RATE_CTRL_SW** and **UL_AMBR_RATE_CTRL_SW** options to their recommended values.
Compatibility Algorithm Switch	**gNodeBParam**.***CompatibilityAlgoSwitch***	Set the **GBR_ADMIT_BY_23501_SW** and **PDU_SESSION_MOD_COMPAT_OPT_SW** options to their recommended values.
Service Differentiation Switch	**NRDUCellAlgoSwitch**.***ServiceDiffSwitch***	Set the **GBR_NONGBR_SCH_OPT_SW** option to the recommended value.
DL Scheduling Priority Coefficient	**gNBDURfspConfig**.***DlSchPriCoefficient***	Set this parameter based on the network plan. Y_{conn} is subject to the settings of this parameter. For details about the impacts of and suggestions on the configuration of Y_{conn}, see **4.2.2 Impacts**.
UL Scheduling Priority Coefficient	**gNBDURfspConfig**.***UlSchPriCoefficient***	Set this parameter based on the network plan. Y is subject to the settings of this parameter. For details about the impacts of and suggestions on the configuration of Y, see **4.2.2 Impacts**.

4.4.1.2 Using MML Commands

Before using MML commands, refer to **4.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

The extended QCI of 128 is used as an example.

```
//Adding QCI-specific bearer configuration for the gNodeB
ADD GNBQCIBEARER: Qci=128;
//Adding 5QI configuration for the gNodeB
ADD GNB5QICONFIG: Nr5qi=128, Qci=128;
//Adding a bearer for an NR DU cell and the MAC parameter group ID and RLC parameter group ID
referenced by the bearer
ADD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=128, MacParamGroupId=9, AmRlcParamGroupId=3,
UmRlcParamGroupId=0;
//Adding a QCI-specific bearer for an NR cell and the PDCP parameter group ID referenced by the bearer
ADD NRCELLQCIBEARER: NrCellId=0, Qci=128, AmPdcpParamGroupId=5, UmPdcpParamGroupId=1;
//Adding an operator-level bearer with the QCI for an NR cell and the PDCP parameter group ID referenced
by the bearer
ADD NRCELLOPQCIBEARER: NrCellId=0, OperatorId=0, Qci=128, RlcMode=AM, AmPdcpParamGroupId=5,
UmPdcpParamGroupId=1;
//Configuring the PDB-based differentiated scheduling policy (NORMAL_SERV is used as an example)
MOD GNBQCIBEARER: Qci=128, PdbDiffSchPolicy=NORMAL_SERV;
//Configuring the maximum scheduling time budget if the PDB-based differentiated scheduling policy is set
to HIGH_RELIAB_SERV
MOD GNBQCIBEARER: Qci=128, PdbDiffSchPolicy=HIGH_RELIAB_SERV, MaxSchTimeBudget=25;
//Enabling the notification control function
MOD GNB5QICONFIG: Nr5qi=128, Nr5qiAlgoSwitch=NOTIF_CTRL_SW-1;
//Configuring the notification control check interval
MOD GNODEBPARAM: NotifCtrlCheckInterval=1;
//Configuring the notification control resource factor
MOD GNBQCIBEARER: Qci=128, NotifCtrlResFactor=100;
//Configuring the QNC Arrange Type parameter (with NORMAL used in this example)
MOD GNBQCIBEARER: Qci=128, QncArrangeType=NORMAL;
//Turning on the PDU Session Resource Modify QCI mapping optimization switch
MOD GNODEBPARAM: CompatibilityAlgoSwitch=PDU_SESSION_MOD_QCI_MAP_OPT_SW-1;
```

Optimization Command Examples

Both standardized and extended QCIs are involved, and the extended QCI of 128 is used as an example.

```
//Configuring the ID of the MAC parameter group referenced by a bearer in an NR DU cell
MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=128, MacParamGroupId=9;
//Configuring the downlink and uplink guaranteed rates for a bearer
MOD GNBDUMACPARAMGROUP: MacParamGroupId=9, DlGuaranteedRate=1000, UlGuaranteedRate=1000;
//Configuring the QCI-specific scheduling priority weighting factors
MOD GNBDUMACPARAMGROUP: MacParamGroupId=9, DlSchPriWeightFactor=700,
UlSchPriWeightFactor=700;
//Configuring scheduling priority weighting factors for bearers with different QCIs or ARPs for different
operators
ADD GNBARP: OperatorId=0, Qci=128, Arp=1, DlSchPriWeightFactor=1000, UlSchPriWeightFactor=1000;
ADD GNBARP: OperatorId=0, Qci=128, Arp=9, DlSchPriWeightFactor=300, UlSchPriWeightFactor=500;
//Modifying the scheduling type, logical channel group ID, and logical channel priority of a bearer
```

```
MOD GNBDUMACPARAMGROUP: MacParamGroupId=9, LogicalChnGrpId=7, LogicalChnPriority=15,
SchedulingType=BASIC_SCH;
//Configuring the RLC mode of a bearer
MOD NRCELLQCIBEARER: NrCellId=0, Qci=128, RlcMode=AM;
//Configuring the ID of the PDCP parameter group referenced by a bearer
MOD NRCELLQCIBEARER: NrCellId=0, Qci=128, AmPdcpParamGroupId=5, UmPdcpParamGroupId=1;
//Configuring the ID of the RLC parameter group referenced by a bearer
MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=128, AmRlcParamGroupId=3, UmRlcParamGroupId=0;
//Configuring QoS attributes mapping a QCI
MOD GNBQCIBEARER: Qci=128, AveragingWindowDur=2000, PacketDelayBudget=600, PriorityLevel=90;
//Configuring the AS Reflective QoS Switch parameter
MOD GNB5QICONFIG: Nr5qi=128, AsReflectiveQosSwitch=ON;
//Subscribing to QCI-related performance counters for NSA UEs
MOD GNBQCIBEARER: Qci=128, CounterObjSubscriptionInd=YES;
//Subscribing to 5QI-related performance counters for SA UEs
MOD GNB5QICONFIG: Nr5qi=128, CounterObjSubscriptionInd=YES;
//Enabling UE-AMBR-based rate limitation (This function is enabled by default; perform this step only when
this function is disabled.)
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
ServiceDiffSwitch=UL_AMBR_RATE_CTRL_SW-1&DL_AMBR_RATE_CTRL_SW-1;
//Turning on the switch for GBR service admission in compliance with 3GPP TS 23.501 and the PDU Session
Modify compatibility optimization switch
MOD GNODEBPARAM:
CompatibilityAlgoSwitch=GBR_ADMIT_BY_23501_SW-1&PDU_SESSION_MOD_COMPAT_OPT_SW-1;
//Enabling scheduling optimization for GBR and non-GBR services
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, ServiceDiffSwitch=GBR_NONGBR_SCH_OPT_SW-1;
//Enabling RFSP-based scheduling priority control (RfspIndex being 3 is used as an example)
ADD GNBDURFSPCONFIG: OperatorId=0, RfspIndex=3, DlSchPriCoefficient=900, UlSchPriCoefficient=900;
```

Deactivation Command Examples

The extended QCI of 128 is used as an example.

```
//Removing QCI-specific bearer configuration from an NR cell
RMV NRCELLQCIBEARER: NrCellId=0, Qci=128;
//Removing operator-level bearer configuration for the QCI from an NR cell
RMV NRCELLOPQCIBEARER: NrCellId=0, OperatorId=0, Qci=128;
//Removing QCI-specific bearer configuration from an NR DU cell
RMV NRDUCELLQCIBEARER: NrDuCellId=0, Qci=128;
//Removing 5QI configuration from the gNodeB
RMV GNB5QICONFIG: Nr5qi=128;
//Removing QCI-specific bearer configuration from the gNodeB
RMV GNBQCIBEARER: Qci=128;
//Disabling UE-AMBR-based rate limitation
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
ServiceDiffSwitch=UL_AMBR_RATE_CTRL_SW-0&DL_AMBR_RATE_CTRL_SW-0;
//Turning off the switch for GBR service admission in compliance with 3GPP TS 23.501 and the PDU Session
Modify compatibility optimization switch
MOD GNODEBPARAM:
CompatibilityAlgoSwitch=GBR_ADMIT_BY_23501_SW-0&PDU_SESSION_MOD_COMPAT_OPT_SW-0;
//Deleting scheduling priority weighting factors for bearers with different QCIs or ARPs for different
operators
RMV GNBARP: OperatorId=0, Qci=128, Arp=1;
RMV GNBARP: OperatorId=0, Qci=128, Arp=9;
//Disabling scheduling optimization for GBR and non-GBR services
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, ServiceDiffSwitch=GBR_NONGBR_SCH_OPT_SW-0;
//Disabling RFSP-based scheduling priority control (RfspIndex being 3 is used as an example)
MOD GNBDURFSPCONFIG: OperatorId=0, RfspIndex=3, DlSchPriCoefficient=1000, UlSchPriCoefficient=1000;
//Disabling downlink PDB-based differentiated scheduling
MOD GNBQCIBEARER: Qci=128, PdbDiffSchPolicy=OFF;
//Disabling the notification control function
MOD GNB5QICONFIG: Nr5qi=128, Nr5qiAlgoSwitch=NOTIF_CTRL_SW-0;
//Turning off the PDU Session Resource Modify QCI mapping optimization switch
MOD GNODEBPARAM: CompatibilityAlgoSwitch=PDU_SESSION_MOD_QCI_MAP_OPT_SW-0;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

QoS Management in the Service Setup Request Phase

NSA Networking

For the Radio QoS Management feature:

1. On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**.
2. In the navigation tree on the left, choose **X2 Interface Trace**.
3. In the **X2 Interface Trace** dialog box that is displayed, confirm that:
   – The value of the *RLC-Config* IE in the SGNB_ADD_REQ_ACK message over the X2 interface is identical to the value of the **NRDUCellQciBearer**.***AmRlcParamGroupId*** or **NRDUCellQciBearer.*UmRlcParamGroupId*** parameter.
   – The value of the *pdcp-Config* IE in the SGNB_ADD_REQ_ACK message over the X2 interface is the same as the value of the **NRCellQciBearer.*AmPdcpParamGroupId*** or **NRCellQciBearer.*UmPdcpParamGroupId*** parameter if the **NRCellOpQciBearer** MO is not configured; or is the same as the value of the **NRCellOpQciBearer**.***AmPdcpParamGroupId*** or **NRCellOpQciBearer**.***UmPdcpParamGroupId*** parameter if the **NRCellOpQciBearer** MO is configured.

   If both of the preceding conditions are met, the Radio QoS Management feature has taken effect.

SA Networking

- For the Radio QoS Management feature:
  a. On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**.
  b. In the navigation tree on the left, choose **Uu Interface Trace**.
  c. In **Uu Interface Trace** dialog box that is displayed, confirm that:
     ■ The value of the *RLC-Config* IE in the RRC_RECONFIGURATION message over the Uu interface is identical to the value of the **NRDUCellQciBearer**.***AmRlcParamGroupId*** or **NRDUCellQciBearer**.***UmRlcParamGroupId*** parameter.
     ■ The value of the *pdcp-Config* IE in the RRC_RECONFIGURATION message over the Uu interface is the same as the value of the **NRCellQciBearer**.***AmPdcpParamGroupId*** or **NRCellQciBearer**.***UmPdcpParamGroupId*** parameter if the **NRCellOpQciBearer** MO is not configured; or is the same as the value of the **NRCellOpQciBearer**.***AmPdcpParamGroupId*** or **NRCellOpQciBearer**.***UmPdcpParamGroupId*** parameter if the **NRCellOpQciBearer** MO is configured.

If both of the preceding conditions are met, the Radio QoS Management feature has taken effect.

- For consistency check between the 5QI-specific service type of a QoS flow and the core-network-delivered one:
  a. On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**.
  b. In the navigation tree on the left, choose **NG Interface Trace**.
  c. In the **NG Interface Trace** dialog box that is displayed, check **PDU Session Resource Setup** > **PDU Session Resource Setup Request Transfer** > **QoS Flow Setup Request Item** > **QoS Flow Level QoS Parameters**. If there is a *GBR QoS Flow Information* IE, the service is a GBR service. Otherwise, the service is a non-GBR service.
  d. Check whether the service type indicated by the *GBR QoS Flow Information* IE is the same as that indicated by the 5QI (see **4.1.1.2 SA Service-to-Bearer Mapping** for more information):
     - If they are consistent, the QoS flow is established successfully.
     - If they are inconsistent, the QoS flow establishment fails.

  In this case, consistency check between the 5QI-specific service type of a QoS flow and the core-network-delivered one has taken effect.
- For PDU Session Resource Modify compatibility optimization:
  a. On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**.
  b. In the navigation tree on the left, choose **NG Interface Trace**.
  c. In the **NG Interface Trace** dialog box that is displayed, check the value of **CHOICE QoS Characteristics** under **PDU Session Resource Setup** > **PDU Session Resource Setup Request Transfer** > **QoS Flow Setup Request Item** > **QoS Flow Level QoS Parameters**, and that under **PDU Session Resource Modify Request** > **PDU Session Resource Modify Request Transfer** > **QoS Flow Add or Modify Request Item** > **QoS Flow Level QoS Parameters**.
  d. If the QoS flow can be modified successfully when the value of **CHOICE QoS Characteristics** is changed from **Non-dynamic 5QI** to **Dynamic 5QI** or from **Dynamic 5QI** to **Non-dynamic 5QI**, the PDU Session Resource Modify compatibility optimization function has taken effect.

QoS Management for Established Services

- For UE-AMBR-based rate limitation for non-GBR services, run the **LST NRDUCELLALGOSWITCH** command and verify that the value of **Service Differentiation Switch** is **UL UE-AMBR-based Control Switch:On** or **DL UE-AMBR-based Control Switch:On** in the command output.
  – In NSA networking, if there is a *SgNB UE Aggregate Maximum Bit Rate* IE in the SGNB ADDITION REQUEST message over the X2 interface, UE-AMBR-based rate limitation for non-GBR services has taken effect. Otherwise, this function does not take effect.
  – In SA networking, if there is a *UE Aggregate Maximum Bit Rate Downlink* IE in the INITIAL CONTEXT SETUP REQUEST or PDU SESSION RESOURCE

SETUP REQUEST message over the NG interface, UE-AMBR-based rate limitation for non-GBR services has taken effect. Otherwise, this function does not take effect.

- For minimum rate guarantee and maximum rate limitation for GBR services:
  – In NSA networking, if there is a *GBR QoS Information* IE in the SGNB ADDITION REQUEST or SGNB MODIFICATION REQUEST message over the X2 interface, both minimum rate guarantee for GBR services and maximum rate limitation for GBR services have taken effect. Otherwise, neither function takes effect.
  – In SA networking, if there is a *GBR QoS Flow Information* IE in the PDU SESSION RESOURCE SETUP REQUEST or PDU SESSION RESOURCE MODIFY REQUEST message over the NG interface, both minimum rate guarantee for GBR services and maximum rate limitation for GBR services have taken effect. Otherwise, both functions do not take effect.
- For ARP- and QCI-based downlink or uplink scheduling priority control:

  You are advised to monitor the single-UE throughput in multi-UE scenarios. It is expected that a smaller scheduling priority weighting factor leads to a lower UE throughput whereas a higher scheduling priority weighting factor leads to a higher UE throughput.
- For RFSP-based downlink or uplink scheduling priority control:

  You are advised to monitor the single-UE throughput in multi-UE scenarios. It is expected that a smaller scheduling priority coefficient leads to a lower UE throughput whereas a higher scheduling priority coefficient leads to a higher UE throughput.
- No valid activation verification method is available for minimum rate guarantee for non-GBR services.
- The following method can be used for observing the activation of downlink PDB-based differentiated scheduling: (You can select any service type for observation. This document uses short video services as an example.)
  a. Configure the 5QI of short video services and set the **gNBQciBearer**.***ServiceType*** parameter to **EMBB_SHORT_VIDEO** for the 5QI.
  b. Set the **gNB5qiConfig**.***CounterObjSubscriptionInd*** parameter to **YES** for the 5QI of short video services.

  After the preceding configurations are completed, you can observe the counters related to the 5QI of short video services to check whether downlink PDB-based differentiated scheduling has been activated. When the value of any of the counters listed in the following table is not 0, downlink PDB-based differentiated scheduling has been activated.

Table 4-12 Cell-level counters for short video service activation verification

Counter ID	Counter Name
1911820520	**N.QosFlow.Max.Cell5QI**
1911820521	**N.QosFlow.Avg.Cell5QI**
1911820488	**N.User.RRCConn.Active.DuCell5QI.Max**

Counter ID	Counter Name
1911820489	**N.User.RRCConn.Active.DuCell5QI.Avg**
1911820510	**N.QosFlow.Est.Att.Cell5QI**
1911820511	**N.QosFlow.Est.Succ.Cell5QI**
1911831584	**N.QosFlow.FailEst.AMF.Cell5QI**
1911831585	**N.QosFlow.FailEst.RNL.Cell5QI**
1911831586	**N.QosFlow.FailEst.TNL.Cell5QI**
1911837955	**N.QosFlow.ReEst.Succ.Cell5QI**
1911820513	**N.QosFlow.AbnormRel.Cell5QI**
1911820514	**N.QosFlow.AbnormRel.AMF.Cell5QI**
1911831588	**N.QosFlow.AbnormRel.RNL.Cell5QI**
1911831589	**N.QosFlow.AbnormRel.TNL.Cell5QI**
1911831587	**N.PDCP.DL.TrfSDU.RxPackets.Cell5QI**
1911831590	**N.PDCP.DL.TrfSDU.TxPacket.Discard.Cell5QI**
1911833638	**N.QosFlow.AbnormRel.HOFailure.Cell5QI**
1911837880	**N.QosFlow.AbnormRel.RNL.RlcReset.Cell5QI**
1911837956	**N.QosFlow.FailEst.Conflict.Cell5QI.PLMN**
1911837881	**N.QosFlow.FailEst.RlcReset.Cell5QI.PLMN**
1911837882	**N.QosFlow.FailEst.UeNoReply.Cell5QI.PLMN**
1911837883	**N.QosFlow.AbnormRel.RNL.RlcReset.Cell5QI.PLMN**
1911837884	**N.QosFlow.FailEst.RlcReset.Cell5QI**
1911837885	**N.QosFlow.FailEst.UeNoReply.Cell5QI**

- The following method can be used for observing the activation of notification control in QoS guarantee for non-scheduling procedure:
  a. On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**.
  b. In the navigation tree on the left of the **Signaling Trace Management** tab page, choose **NR > Application Layer >NG Interface Trace**.
  c. If the PDU SESSION RESOURCE NOTIFY message over the NG interface contains the *Notification Cause* IE, the notification control function has taken effect.

4.4.3 Network Monitoring

NSA Networking

In NSA networking, monitor the following performance counters.

Counter ID	Counter Name
1911817047	**N.ThpVol.DL.QCI**
1911817050	**N.ThpVol.DL.LastSlot.QCI**
1911817051	**N.ThpTime.DL.RmvLastSlot.QCI**

The average downlink throughput of UEs carried on bearers with a specific QCI is calculated as follows: (**N.ThpVol.DL.QCI – N.ThpVol.DL.LastSlot.QCI**)/ **N.ThpTime.DL.RmvLastSlot.QCI**. Currently, there is no counter available for monitoring the average uplink throughput of UEs carried on bearers with a specific QCI.

By default, only subscription to QCI 9-related counters is available. To subscribe to the performance counters of another QCI, set the **gNBQciBearer**.***CounterObjSubscriptionInd*** parameter corresponding to this QCI to **YES** before network monitoring.

SA Networking

In SA networking, monitor the following performance counters.

Counter ID	Counter Name
1911820485	**N.ThpVol.DL.DuCell5QI**
1911820483	**N.ThpVol.DL.LastSlot.DuCell5QI**
1911820487	**N.ThpTime.DL.RmvLastSlot.DuCell5QI**
1911822887	**N.RLC.DL.TrfSDU.PktUuTrans.Loss.DuCell5QI**
1911822886	**N.RLC.DL.TrfSDU.RxPackets.DuCell5QI**
1911822889	**N.PDCP.UL.TrfSDU.RxPacket.Loss.Cell5QI**
1911822888	**N.PDCP.UL.TrfPDU.RxPackets.Cell5QI**
1911825119	**N.ThpVol.UL.DuCell5QI**

The average downlink throughput of UEs carried on bearers with a specific 5QI is calculated as follows: (**N.ThpVol.DL.DuCell5QI – N.ThpVol.DL.LastSlot.DuCell5QI**)/**N.ThpTime.DL.RmvLastSlot.DuCell5QI**. Currently, there is no counter available for monitoring the average uplink throughput of UEs carried on bearers with a specific 5QI.

The average downlink RLC packet loss rate of UEs carried on bearers with a specific 5QI is calculated as follows: **N.RLC.DL.TrfSDU.PktUuTrans.Loss.DuCell5QI**/ **N.RLC.DL.TrfSDU.RxPackets.DuCell5QI**. Currently, there is no counter available for monitoring the average uplink RLC packet loss rate of UEs carried on bearers with a specific 5QI.

The average uplink PDCP packet loss rate of UEs carried on bearers with a specific 5QI is calculated as follows: **N.PDCP.UL.TrfSDU.RxPacket.Loss.Cell5QI**/ **N.PDCP.UL.TrfPDU.RxPackets.Cell5QI**. Currently, there is no counter available for monitoring the average downlink PDCP packet loss rate of UEs carried on bearers with a specific 5QI.

By default, only subscription to 5QI 9-related counters is available. To subscribe to the performance counters of another 5QI, set the **gNB5qiConfig**.***CounterObjSubscriptionInd*** parameter corresponding to this 5QI to **YES** before network monitoring.

5 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

Step 1 Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

6 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

7 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

8 Reference Documents

- *Transmission Resource Management*
- *Uplink Scheduling*
- *Flexible User Steering*
- *FWA*
- *URLLC*
- *QCI-based Resource Control*
- *Layered QoS*
- *Service-differentiated Experience Guarantee*
- *Ultra Uplink (High-Frequency TDD)*
- *Turbo Uplink (Low-Frequency TDD)*
- *CA Experience Improvement*
- *QoS Management* in *eRAN Feature Documentation*
- 3GPP TS 23.203: "Policy and charging control architecture"
- 3GPP TS 23.501: "System Architecture for the 5G System"
- 3GPP TS 38.300: "NR; NR and NG-RAN Overall Description"
- 3GPP TS 38.413: "NR; NG Application Protocol (NGAP)"
- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- *Technical Specifications* in *3900 & 5900 Series Base Station Product Documentation*